Nonuniform weak-coupling expansion in scalar scattering from near-extremal Kerr

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We study weakly nonlinear scattering of a complex scalar field from a Kerr black hole with $U(1)$-invariant interactions $|\Phi|^{2n}\Phi$. The spin-weight-zero Teukolsky equation is solved by an endpoint-inclusive, two-domain Chebyshev method, and the first-order field is constructed with a radial Green function. Transmission and reflection corrections are defined as derivatives of the physical energy fluxes at fixed incident flux. For axisymmetric modes, matched static and far-zone analysis gives $T^{(0)}_{\ell 0}, T^{(1)}_{\ell 0} \propto (M\omega)^{2\ell+2}$ and a finite, spin-independent ratio $\chi_{H,\ell 0} = T^{(1)}/T^{(0)}$. At the amplitude level, the apparent $(M\omega)^{-2\ell-2}$ growth of the unit-horizon Green coefficient is removed by fixed-incident normalization: the relative horizon amplitude tends to a finite complex constant whose real and imaginary parts give the absorption and phase responses, respectively. The monopole obeys $M^{2n-2}\chi^{(n)}_{H,00} \sim c_n (M\omega)^{2n-2}$ for every positive integer n, with $\chi^{(1)}_{H,00}(0) = \ln 2/\pi$; independent $n = 2, 3$ calculations recover the analytic powers and coefficients. Thus the cubic interaction is infrared marginal in this channel, whereas the higher local powers are irrelevant.

For nonaxisymmetric modes, both horizon fluxes vanish at the superradiant threshold $p = \omega - m\Omega_H = 0$, while χ_H remains finite and positive at fixed subextremal spin. Near extremality, principal complex-conformal-weight modes exhibit an inverse-surface-gravity envelope and collapse approximately in $(p/\kappa, M\kappa\chi_H)$; real-weight controls do not. Extending the calculation through $a/M = 0.9999984375$ for $n = 2, 3$ supports the order-locked hierarchy $M^{2n-2}\chi^{(n)}_H = (M\kappa)^{-n}\mathcal{F}_{n,\ell}$ in the principal $(2,2)$ and $(3,3)$ channels, with bounded finite-window modulation rather than an exact pure power. Integration over a fixed p/κ interval softens this pointwise hierarchy to κ^{1-n}: the cubic spectral weight is finite at leading power, while the higher weights remain enhanced. Dense profiles also show that increasing n moves the response peak toward the absorptive side and narrows it.

Matched near-NHEK theory yields the factorization $M^{2n-2}\chi^{(n)}_H = \mathcal{I}^n_H \mathcal{K}_{n,h}$, separating the analytically known linear intensity from an unevaluated nonlinear Green kernel. It explains why no conformal-weight-only law follows for supplementary modes and why the principal weak-coupling limit is nonuniform: successive local operators gain an additional $|\mathcal{A}_{\rm in}|^2/(M\kappa)$ in their relative control parameter and can reverse their moderate-spin ordering. Spectral tails, off-grid residuals, paired resolutions, Wronskians, and flux balance provide independent checks. The result is a prescribed scalar response on fixed Kerr, not a metric perturbation or gravitational self-force observable.

I. INTRODUCTION

We use units $G = c = \hbar = 1$, metric signature $(-,+,+,+)$, and set $M = 1$ in the numerical results unless stated otherwise. The Fourier convention is fixed by the mode dependence $e^{-i\omega t + im\phi}$ used below.

Scattering is a foundational probe of black-hole spacetimes. The linear problem has been studied extensively for scalar, electromagnetic, and gravitational waves in Schwarzschild and Kerr backgrounds, using partial-wave, complex-angular-momentum, and related frequency-domain methods [1–3, 5–7]. In a standard scattering experiment one sends in a monochromatic mode from past null infinity and compares the incident flux with the reflected flux at future null infinity and the absorbed flux at the future event horizon. The resulting transmission and reflection coefficients are natural complements to quasinormal-mode spectroscopy [8]. At the level of finite-energy wave packets, the scalar wave equation on every subextremal Kerr exterior also admits a bounded, asymptotically complete scattering map that includes superradiant reflection [4]. This point has become sharper because quasinormal spectra can be sensitive to small deformations of the effective potential [9–11], while scattering data, graybody factors, and related response functions can remain stable observables [12–16].

The nonlinear scattering literature contains several nearby but inequivalent problems. Characteristic evolutions of an incident gravitational pulse have resolved mode coupling, Bondi flux, and the map from incoming to outgoing radiation in Schwarzschild, first in fully nonlinear numerical relativity and more recently with domains extending to null infinity [17–19]. Simulations of the Einstein–Klein–Gordon system and of vacuum gravitational waves have further exhibited harmonic generation, spectral broadening, and spatial focusing [37]. These are time-dependent, self-gravitating scattering experiments; their natural outputs are waveforms, Bondi quantities, and generated frequencies rather than a stationary first derivative of a scalar graybody factor.

Self-interacting scalar fields on Kerr have instead been studied mainly as massive quasi-bound clouds. Time evolutions of an axion cloud have found bosenova collapse, while perturbative and occupation-number descriptions have analyzed self-interaction-induced level transfer, scalar emission, and possible saturation of superradiant growth [20, 22–25]. Nonlinear Q-clouds provide a complementary stationary test-field construction with quartic and hexic interactions [21]. Those calcula-

tions impose decay at infinity and use a complex bound-state frequency or exact synchronization, $\omega = m\Omega_H$, with no incident flux from infinity. The present problem instead uses a massless field, a real driving frequency, and scattering boundary data. It therefore measures an instantaneous open-channel response, not a cloud growth rate, a saturation amplitude, or a stationary scalar condensate.

A complementary fixed-background construction keeps the flux definition of scattering coefficients explicit: one introduces a weak scalar self-interaction, solves the linear homogeneous problem, uses the nonlinear source to construct a first-order Green-function correction, and defines nonlinear scattering coefficients from the interference terms in the energy flux. This is the key point: the first-order field is not a new independent linear scattering experiment; it changes the physical boundary fluxes by interfering with the zeroth-order wave.

The scalar model is not a gravitational self-force calculation, but its bookkeeping is closely related to the sourced perturbative hierarchies used in extreme-mass-ratio inspiral (EMRI) theory: a lower-order field generates a higher-order source, mode coupling must be projected consistently, and observable fluxes provide an important global check [26–28]. We use this analogy only to motivate the workflow; no claim about a gravitational waveform or a self-force observable is made here. Frequency-domain calculations for a scalar charge on Kerr, generic bound Kerr geodesics, and the associated regularization parameters make the additional worldline and singular-field structures explicit [29, 31, 32]. Effective-source and extended-particular-solution methods provide a related scalar prototype for controlling frequency-domain reconstruction, but they retain a singular worldline source and regularization step absent here [30]. Recent second-order self-force and metric-reconstruction work further shows why those structures cannot be replaced by the finite cubic source used here [33, 34]. Recent effective-source constructions make the distinction concrete for both generic first-order orbits and second-order quasicircular calculations: the residual equation carries puncture data, worldline information, and slow-time source terms that are absent from the present stationary scalar problem [35, 36].

The distinction is also important when comparing this scalar prototype with nonlinear Teukolsky calculations. In the gravitational problem the second-order source contains tetrad and metric-perturbation terms and must be combined with gauge completion and, for self-force applications, a singular-field subtraction [56–58]. Our source is instead a smooth local cubic functional of one prescribed scalar mode. We therefore use the EMRI and second-order literature to delimit the interpretation of the calculation, not to identify the scalar coefficient with a gravitational self-force observable.

Here we implement this construction for the scalar Teukolsky equation in Kerr spacetime with spin weight $s = 0$, within the standard analytic black-hole perturbation framework [52]. We retain the usual incident, reflected, and horizon-flux observables while deriving their weakly nonlinear Kerr corrections. Rotation produces three changes. First, the horizon flux is proportional to $p = \omega - m\Omega_H$, which is the origin of superradiant sign changes [40, 41]. Second, spherical harmonics are replaced by spheroidal harmonics. Third, the covariant scalar source contains $\Sigma = r^2 + a^2\cos^2\theta$, so a single incident mode sources several target spheroidal channels.

The near-extremal superradiant bound is a distinguished frequency scale even in the linear theory. Near-superradiant scalar absorption in near-extremal Kerr is reproduced by a finite-temperature two-dimensional conformal correlator, with the right-moving temperature proportional to the surface gravity and the scaled right-moving frequency proportional to p/κ [42–44]. In exactly extremal Kerr the same bound becomes a branch point of the frequency-domain Green function, near-extremal mode families condense toward it, and the throat exhibits an emergent self-similarity [47–50]. These linear results motivate using κ rather than Ω_H alone to resolve the nonlinear response.

The nonlinear observable studied here is not itself a Kerr/CFT two-point function. The equation-level vertex $|\Phi|^{2n}\Phi$ descends from a bulk $|\Phi|^{2n+2}$ interaction and corresponds schematically to a $(2n+2)$-point response in the boundary description. Existing Kerr/CFT higher-point and source-driven calculations establish the relevant holographic framework [45, 46], but do not give the fixed-incident, asymptotically flat nonlinear flux susceptibility defined below. We therefore use Kerr/CFT only to organize the throat scaling and interpret the conformal-weight split; the sign, coefficient, and complete detuning dependence are computed from the nonlinear radial Green function. Our calculation remains strictly subextremal and does not insert an analytic near-horizon formula into that Green integral. Weakly nonlinear gravitational work has also derived a dynamical system for near-extremal zero-damped Kerr modes and found a long-lived inverse-cascade equilibrium [38, 39]. That complex-frequency, second-order Einstein problem motivates sensitivity to the throat scale, but does not predict the real-frequency scalar coefficient calculated here.

The paper follows the physical order of the construction, from the field equation and boundary fluxes to the Green function and numerical diagnostics. Section II defines the nonlinear scattering problem and the flux coefficients. Section III describes the pseudospectral homogeneous solver, the Green-function construction, the balance law, the accuracy diagnostics, and the low- and high-frequency diagnostics. Section IV first gives Kerr nonlinear transmission curves for $\ell = 0, 1, 2$, their resonance fits, and the analytic axisymmetric infrared laws. It then turns to nonaxisymmetric scattering, separates the threshold and near-extremal limits, and tests the principal/supplementary split and the operator-order hierarchy. The initial frequency sweep is deliberately restricted to $a = 0.5$ and $m = 0$ so that radial and

spheroidal effects are isolated before the superradiant horizon factor is introduced; these data are not mixed with the later cross-spin survey.

The contributions of this paper are fourfold. First, we derive the Kerr fixed-incident normalization of the weak cubic response, including the horizon factor p and the Σ-weighted spheroidal source projection, so that the reported quantities are energy-flux derivatives rather than unnormalized Green amplitudes. Second, we implement the endpoint-inclusive two-domain Chebyshev construction and combine spectral tails, field-dependent radial residuals, Wronskian constancy, and flux balance into an explicit acceptance protocol. Third, we use that protocol to impose a common flux-balance acceptance gate before plotting or fitting, while exposing the off-diagonal target-channel structure and deriving the absolute infrared flux laws. The monopole classifies the full $U(1)$-invariant family $|\Phi|^{2n}\Phi$: the cubic equation is marginal, whereas every $n > 1$ response vanishes as ω^{2n-2}. Fourth, we resolve a distinct near-superradiant scaling pattern. The principal, complex-conformal-weight modes have a numerical $1/\kappa$ envelope for the cubic interaction and collapse in p/κ and $M\kappa\chi_H$. For $n = 2, 3$, the data support $M^{2n-2}\chi_H^{(n)} = (M\kappa)^{-n}\mathcal{F}^{\rm num}_{n,\ell}$ with a bounded, slowly varying factor over the resolved paths. We trace this operator-order hierarchy to fixed-flux throat intensity and separate the exact linear matching factor from the unevaluated nonlinear Green kernel. This establishes both the sign and the geometric scale of the first nonlinear correction. A fixed throat-frequency integration softens the principal hierarchy by one power but does not remove it for $n > 1$, and dense peak scans resolve a shift toward the absorptive side with increasing operator order. All claims remain restricted to the prescribed scalar response problem defined above.

II. NONLINEAR SCATTERING PROBLEM IN KERR SPACETIME

In Boyer-Lindquist coordinates the Kerr metric is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1-\frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\mathrm{d}t\,\mathrm{d}\phi \\ &+ \frac{\Sigma}{\Delta}\mathrm{d}r^2 + \Sigma\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\mathrm{d}\phi^2, \end{aligned} \tag{1}$$

where

$$\Sigma = r^2 + a^2\cos^2\theta, \quad \Delta = (r-r_+)(r-r_-), \tag{2}$$

$$A = (r^2+a^2)^2 - a^2\Delta\sin^2\theta. \tag{3}$$

The horizon radii and angular velocity are

$$r_\pm = M \pm \sqrt{M^2-a^2}, \qquad \Omega_H = \frac{a}{r_+^2+a^2}. \tag{4}$$

We consider a weak cubic self-interaction

$$\Box\Phi + \epsilon|\Phi|^2\Phi = 0, \qquad \epsilon \in \mathbb{R}, \quad |\epsilon| \ll 1, \tag{5}$$

Here ϵ is a signed perturbative coefficient. With the action convention in Eq. (7), a conventional positive quartic potential $V = \lambda|\Phi|^4/2$ corresponds to $\epsilon = -\lambda$; therefore the sign statements below always refer to Eq. (5), not to a separate stability assumption about the scalar potential. In four spacetime dimensions the canonical scalar has dimensions $[\Phi] = M^{-1}$, so ϵ is dimensionless in the action convention below. For a field amplitude that is not absorbed into the unit-horizon normalization, a useful local control parameter is

$$\chi_{\rm nl} = |\epsilon| M^2 \max_{\mathcal{D}} |\Phi^{(0)}|^2 \ll 1, \tag{6}$$

where $\mathcal{D}$ is the radial and angular region that dominates the cubic source projection. The reported $T^{(1)}$ and $R^{(1)}$ are coefficients in the derivative of the fixed-incident flux with respect to ϵ at $\epsilon = 0$; a finite-coupling application must additionally check that $|\epsilon||\mathcal{A}_{\rm in}|^2|T^{(1)}| \ll |T^{(0)}|$ and the analogous reflected-flux condition. We do not interpret the present frequency-domain calculation as a test of nonlinear saturation or of the expansion beyond this perturbative regime. This equation follows from the fixed-background scalar action

$$S_\Phi = \int \mathrm{d}^4x\,\sqrt{-g}\left[-g^{\mu\nu}\nabla_\mu\Phi^*\nabla_\nu\Phi + \frac{\epsilon}{2}|\Phi|^4\right]. \tag{7}$$

The stress tensor used below is the Hilbert stress tensor of the same effective field theory,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_\Phi}{\delta g^{\mu\nu}}. \tag{8}$$

For later use, the explicit tensor and the associated stationary Killing energy flux are

$$T_{\mu\nu} = \nabla_\mu\Phi^*\nabla_\nu\Phi \tag{9}$$

$$+ \nabla_\nu\Phi^*\nabla_\mu\Phi + g_{\mu\nu}\left[-\nabla_\alpha\Phi^*\nabla^\alpha\Phi + \frac{\epsilon}{2}|\Phi|^4\right], \tag{10}$$

$$j_E^\mu = -T^\mu{}_\nu\xi^\nu = -T^\mu{}_t, \qquad \xi = \partial_t,$$

$$\mathcal{F}_E^{\rm out}(r) = \int_0^{2\pi}\mathrm{d}\phi\int_0^\pi \mathrm{d}\theta\,\sqrt{-g}\,j_E^r, \tag{11}$$

$$\mathcal{F}_E^H = -\lim_{r\to r_+}\mathcal{F}_E^{\rm out}(r). \tag{12}$$

The reversed horizon orientation in the last definition makes positive $\mathcal{F}_E^H$ correspond to absorption when $p > 0$. The quartic term is proportional to $\delta^r{}_t$ in $T^r{}_t$ and therefore does not itself carry radial flux; the nonlinear boundary correction comes from the interference of the zeroth- and first-order kinetic terms. We now expand the field as

$$\Phi = \Phi^{(0)} + \epsilon\Phi^{(1)} + \mathcal{O}(\epsilon^2). \tag{13}$$

The zeroth-order incident wave is a single scalar spheroidal mode,

$$\Phi^{(0)} = e^{-\mathrm{i}\omega t+\mathrm{i}m\phi} S_{\ell m}(\theta; a\omega) R_{\ell m\omega}(r), \tag{14}$$

with angular normalization

$$\int |S_{\ell m}|^2 \mathrm{d}\Omega = 1. \tag{15}$$

The scalar spheroidal harmonic obeys

$$\left[\frac{1}{\sin\theta}\partial_\theta \sin\theta \partial_\theta + a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta} + A_{\ell m}\right] S_{\ell m} = 0. \tag{16}$$

The radial Teukolsky equation for $s = 0$ is [59]

$$\mathcal{L}_{\ell m\omega} R \equiv \frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta \frac{\mathrm{d}R}{\mathrm{d}r}\right) + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right) R = 0, \tag{17}$$

where

$$K = (r^2+a^2)\omega - am, \qquad \lambda_{\ell m\omega} = A_{\ell m} + a^2\omega^2 - 2am\omega. \tag{18}$$

The endpoint normalization used here can be related to the analytic MST solutions, which provide independent low-frequency connection formulae [51]. The numerical results below are assessed using internal spectral convergence and flux-balance diagnostics. The tortoise coordinate is defined by

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{r^2+a^2}{\Delta}, \tag{19}$$

and the horizon wave number is

$$p = \omega - m\Omega_H. \tag{20}$$

The ingoing solution is normalized to unit amplitude at the future horizon,

$$R^{\mathrm{in}}_{\ell m\omega} \sim e^{-\mathrm{i}pr_*}, \qquad r \to r_+, \tag{21}$$

and at infinity it has the form

$$R^{\mathrm{in}}_{\ell m\omega} \sim B^{\mathrm{inc}} \frac{e^{-\mathrm{i}\omega r_*}}{r} + B^{\mathrm{ref}} \frac{e^{+\mathrm{i}\omega r_*}}{r}, \qquad r \to \infty. \tag{22}$$

With this convention the linear flux coefficients are

$$T^{(0)} = \frac{p(r_+^2+a^2)}{\omega |B^{\mathrm{inc}}|^2}, \qquad R^{(0)} = \frac{|B^{\mathrm{ref}}|^2}{|B^{\mathrm{inc}}|^2}. \tag{23}$$

For clarity, these coefficients are energy-flux ratios rather than merely amplitude ratios. Let $\xi = \partial_t$ be the stationary Killing field and let $T_{\mu\nu}$ be the scalar stress tensor defined in Eq. (8). The outward-oriented Killing energy current is $j^\mu_E = -T^\mu{}_t$, so at a surface of constant r we use $\mathcal{F}^{\mathrm{out}}_E = \int j^r_E \sqrt{-g}\,\mathrm{d}\theta\mathrm{d}\phi$. When quoting absorption by the future horizon, the horizon boundary orientation is reversed, so that positive absorption corresponds to positive $\mathcal{F}^H_E = -\mathcal{F}^{\mathrm{out}}_E|_{r=r_+}$ for $p > 0$. For a single Fourier mode, the energy current is the mode frequency times the radial Klein–Gordon current, up to one common positive normalization $\mathcal{N}$. Consequently,

$$\begin{aligned} \mathcal{F}^{\mathrm{in}}_{E,\infty} &= \mathcal{N}\omega^2 |B^{\mathrm{inc}}|^2, \\ \mathcal{F}^{\mathrm{ref}}_{E,\infty} &= \mathcal{N}\omega^2 |B^{\mathrm{ref}}|^2, \\ \mathcal{F}_{E,H} &= \mathcal{N}\omega p (r_+^2+a^2). \end{aligned} \tag{24}$$

Their ratios give Eq. (23), including the sign of the horizon flux when $p < 0$. The self-interaction contributes a term proportional to $\delta^r{}_t$ in the mixed stress tensor and therefore does not add a radial boundary flux; at first order, the nonlinear flux correction is the interference term between $\Phi^{(0)}$ and $\Phi^{(1)}$ derived below.

For real ω the conserved radial current gives

$$T^{(0)} + R^{(0)} = 1. \tag{25}$$

At first order,

$$\Box \Phi^{(1)} = -|\Phi^{(0)}|^2 \Phi^{(0)}. \tag{26}$$

Because the scalar is complex and the interaction is $U(1)$ invariant, $|e^{-i\omega t+im\phi}|^2 e^{-i\omega t+im\phi}$ has the same (ω, m) dependence as the incident mode. We therefore expand

$$\Phi^{(1)} = e^{-\mathrm{i}\omega t+\mathrm{i}m\phi} \sum_{\ell' \geq |m|} S_{\ell' m}(\theta; a\omega) R^{(1)}_{\ell'|\ell m\omega}(r). \tag{27}$$

After multiplying by Σ and projecting onto a target spheroidal harmonic, the radial source is

$$\mathcal{L}_{\ell' m\omega} R^{(1)}_{\ell'|\ell m\omega} = -Q_{\ell'\ell m}(r), \tag{28}$$

where

$$Q_{\ell'\ell m}(r) = \left[r^2 C^{(0)}_{\ell'\ell m} + a^2 C^{(2)}_{\ell'\ell m}\right] |R^{\mathrm{in}}_{\ell m\omega}|^2 R^{\mathrm{in}}_{\ell m\omega}, \tag{29}$$

and

$$C^{(j)}_{\ell'\ell m} = \int \mathrm{d}\Omega\, \cos^j\theta\, S^*_{\ell' m} |S_{\ell m}|^2 S_{\ell m}, \qquad j = 0, 2. \tag{30}$$

The self channel $\ell' = \ell$ is the only target channel that interferes with the linear outgoing wave in the total energy flux at order ϵ.

III. METHODS

A. Pseudospectral method for homogeneous solutions

The homogeneous radial equation is solved in the compact coordinate

$$z = \frac{r_+}{r}, \qquad z = 0 \text{ at infinity}, \qquad z = 1 \text{ at the horizon}. \tag{31}$$

The oscillatory endpoint phases are peeled off before Chebyshev collocation. This endpoint-first organization is also consistent with independent spectral black-hole perturbation work, where the asymptotic behavior is factored before the Chebyshev representation is truncated [53]:

$$R_{\rm in} = e^{-{\rm i}pr_*}u_{\rm in}(z), \tag{32}$$

$$R_{\rm down} = \frac{e^{-{\rm i}\omega r_*}}{r}u_{\rm down}(z), \tag{33}$$

$$R_{\rm up} = \frac{e^{+{\rm i}\omega r_*}}{r}u_{\rm up}(z). \tag{34}$$

The functions $u_{\rm in}$, $u_{\rm down}$, and $u_{\rm up}$ are regular on their respective subdomains that the degenerate endpoint rows of the transformed ODE can be used directly as boundary regularity conditions. The outer and inner domains are matched at a finite radius $r_{\rm m}$ by continuity of R and $\mathrm{d}R/\mathrm{d}r$.

More explicitly, after the phase peeling the collocation equation has the form

$$B_2(z)u_{,zz} + B_1(z)u_{,z} + B_0(z)u = 0. \tag{35}$$

These coefficients follow directly without a further field redefinition. Writing $R = f(r)u[z(r)]$, $g = \mathrm{d}z/\mathrm{d}r = -z^2/r_+$, and $V = K^2/\Delta - \lambda_{\ell m\omega}$, one obtains

$$\begin{aligned} B_2 &= \Delta g^2, \\ B_1 &= \Delta\left(g_{,r} + 2g\frac{f_{,r}}{f}\right) + \Delta_{,r}g, \\ B_0 &= \Delta\frac{f_{,rr}}{f} + \Delta_{,r}\frac{f_{,r}}{f} + V, \end{aligned} \tag{36}$$

where $f = e^{-ipr_*}$ for the inner ingoing branch and $f = r^{-1}e^{\mp i\omega r_*}$ for the outer down/up branches. The expressions used in the code are the analytic endpoint limits of Eq. (36), so no finite-difference evaluation of a singular quotient is required. At the exact endpoint of each branch B_2 vanishes, so the endpoint row is the regular-singular relation $B_1u_{,z} + B_0u = 0$. The neighboring row is replaced by the normalization $u_{\rm down}(0) = u_{\rm up}(0) = 1$ on the outer domain or $u_{\rm in}(1) = 1$ on the inner domain. Thus the horizon is included as an exact collocation endpoint; no horizon truncation is introduced. Each domain uses $N+1$ Chebyshev points and a dense complex linear solve. For $|M\omega| < 0.1$ we use the mapped nodes

$$z(y) = z_L + (z_R - z_L)\frac{\sinh[\kappa(1-y)/2]}{\sinh\kappa}, \qquad -1 \le y \le 1, \tag{37}$$

with $\kappa_{\rm out} = |\log(2M|\omega|)|$ and $\kappa_{\rm in} = \kappa_{\rm out}/2$; otherwise the map is linear. This redistributes the same endpoint-inclusive Chebyshev grid toward the outer end of each subdomain and does not move either physical boundary. The low-frequency and resonance-region production runs use $N = 240$ on both domains, $r_{\rm m} = 40M$, Gauss order $q = 192$ for the compact inner integral, and relative tolerance 10^{-11} for the oscillatory phase-channel tail. The high-frequency fits use 25 separately recomputed points at the same N and $r_{\rm m}$ but with $q = 2048$. The larger order is required because the compact horizon-side Green integral remains oscillatory after the homogeneous endpoint phases have been peeled. The tail-order audit compares $q = 1792$ with $q = 2048$ and independently lowers the radial order to $N = 224$. The low-frequency runs use the automatic map in Eq. (37); all reported results are obtained in IEEE double precision and are checked by the order, matching-radius, quadrature, Wronskian, and flux scans reported below.

The angular eigenvalue is computed by diagonalizing the truncated spherical harmonic representation of $\ell(\ell+1) - (a\omega)^2\cos^2\theta$, retaining sixteen modes above the requested ℓ. The source projectors are evaluated with normalized spheroidal harmonics and $q_\theta = 256$ point Gauss–Legendre quadrature. The separate basis and quadrature audit in Table XII compares these active settings with 24 extra modes and $q_\theta = 512$. For the radial Green integrals, the compact inner part is integrated in z with the Jacobian $\mathrm{d}r = r_+/z^2\,\mathrm{d}z$, while the outer part is decomposed into its finite phase channels in r_* before oscillatory quadrature.

The direct Teukolsky potential is long ranged, so an alternative numerical route is to transform the equation to a short-ranged generalized Sasaki–Nakamura equation [54, 55]. We retain the direct Teukolsky variable here because it keeps the covariant Σ-weighted source and the endpoint normalization in the same representation. The GSN formulation is consequently a useful independent route for future high- frequency or extended-source checks, but it is not used to define any of the production values reported below.

The scattering relation used in the code is the usual in/down/up decomposition,

$$R_{\rm in} = B^{\rm inc}R_{\rm down} + B^{\rm ref}R_{\rm up}. \tag{38}$$

At $r = r_{\rm m}$ the amplitudes are obtained from

$$\begin{pmatrix} R_{\rm down} & R_{\rm up} \\ R'_{\rm down} & R'_{\rm up} \end{pmatrix}_{r_{\rm m}} \begin{pmatrix} B^{\rm inc} \\ B^{\rm ref} \end{pmatrix} = \begin{pmatrix} R_{\rm in} \\ R'_{\rm in} \end{pmatrix}_{r_{\rm m}}. \tag{39}$$

Solving this two-by-two matching system gives $B^{\rm inc}$ and $B^{\rm ref}$ and therefore the linear coefficients in Eq. (23). The residual $|T^{(0)} + R^{(0)} - 1|$ is used as the first check of the homogeneous solve.

B. Nonlinear scattering coefficients

1. *The nonlinear balance law*

Let $R^{(\ell)}_{\rm up}$ denote the target-channel homogeneous solution that is purely outgoing at infinity. The Green-function solution in the self channel has outgoing amplitude at infinity and ingoing amplitude at the horizon. The conserved Wronskian is

$$W_\ell = \Delta\left(R^{\rm in}_{\ell m\omega}\frac{\mathrm{d}R^{\rm up}_{\ell m\omega}}{\mathrm{d}r} - R^{\rm up}_{\ell m\omega}\frac{\mathrm{d}R^{\rm in}_{\ell m\omega}}{\mathrm{d}r}\right), \tag{40}$$

which is independent of r for two homogeneous solutions of Eq. (17). The Green-function amplitudes are

$$A^{(1)}_{\rm ref} = -\frac{1}{W_\ell}\int_{r_+}^{\infty} Q_{\ell\ell m}(r) R^{\rm in}_{\ell m\omega}(r)\,{\rm d}r, \tag{41}$$

$$A^{(1)}_{H} = -\frac{1}{W_\ell}\int_{r_+}^{\infty} Q_{\ell\ell m}(r) R^{\rm up}_{\ell m\omega}(r)\,{\rm d}r, \tag{42}$$

where the minus sign follows from $\mathcal{L}_{\ell m\omega} R^{(1)} = -Q_{\ell\ell m}$. In the numerical implementation the Wronskian is evaluated from the matched homogeneous solutions and checked against its constancy. The amplitudes $A^{(1)}_{\rm ref}$ and $A^{(1)}_{H}$ are quoted in the unit-horizon normalization of Eq. (21). To form scattering coefficients one must instead hold the incident wave fixed. If the zeroth-order field is rescaled by $\alpha = 1/B^{\rm inc}$, the cubic source scales as $|\alpha|^2\alpha$. Therefore the fixed-incident first-order boundary amplitudes are

$$A^{(1)}_{\rm ref,fix} = \frac{A^{(1)}_{\rm ref}}{|B^{\rm inc}|^2 B^{\rm inc}}, \qquad A^{(1)}_{H,\rm fix} = \frac{A^{(1)}_{H}}{|B^{\rm inc}|^2 B^{\rm inc}}. \tag{43}$$

The zeroth-order fixed-incident horizon amplitude is $1/B^{\rm inc}$. It is useful to retain the complex relative correction

$$\rho_H(\omega) \equiv \frac{A^{(1)}_{H,\rm fix}}{1/B^{\rm inc}} = \frac{A^{(1)}_{H}}{|B^{\rm inc}|^2}. \tag{44}$$

For unit incident amplitude the total horizon coefficient is therefore $(1/B^{\rm inc})[1+\epsilon\rho_H+\mathcal{O}(\epsilon^2)]$. To first order, ${\rm Re}\,\rho_H$ changes its modulus whereas ${\rm Im}\,\rho_H$ shifts its phase. Only the former enters an energy flux at this order. Expanding the horizon and infinity fluxes of the fixed-incident field then gives

$$T^{(1)} = \frac{2p(r_+^2+a^2)}{\omega|B^{\rm inc}|^4}\,{\rm Re}\,A^{(1)}_{H}, \tag{45}$$

$$R^{(1)} = \frac{2\,{\rm Re}(\overline{B^{\rm ref}}\,A^{(1)}_{\rm ref})}{|B^{\rm inc}|^4}. \tag{46}$$

For $p \neq 0$, division by the linear horizon coefficient removes the explicit superradiant factor and defines the threshold-regular relative response

$$\chi_H(\omega) \equiv \frac{T^{(1)}}{T^{(0)}} = \frac{2\,{\rm Re}\,A^{(1)}_{H}}{|B^{\rm inc}|^2} = 2\,{\rm Re}\,\rho_H. \tag{47}$$

The right-hand side remains directly evaluable as $p \to 0$. Provided the real-frequency homogeneous amplitudes and the projected Green integral are smooth there,

$$\begin{aligned} T^{(0)} &= p\tau_0 + \mathcal{O}(p^2),\\ T^{(1)} &= p\tau_1 + \mathcal{O}(p^2),\\ \chi_H^\star &= \lim_{p\to0}\chi_H = \frac{\tau_1}{\tau_0}. \end{aligned} \tag{48}$$

This form avoids a numerical 0/0 operation and separates the physical vanishing of the horizon flux from cancellation in the reflected interference term. In the superradiant regime the amplification factor $Z \equiv R-1 = -T$ consequently obeys

$$Z = Z^{(0)}\left[1+\epsilon|\mathcal{A}_{\rm in}|^2\chi_H\right] + \mathcal{O}(\epsilon^2|\mathcal{A}_{\rm in}|^4), \qquad p<0. \tag{49}$$

Equivalently, with the incident energy flux held fixed,

$$T^{(1)} = \frac{1}{\mathcal{F}^{\rm in}_{E,\infty}}\frac{\partial\mathcal{F}^{H}_{E}}{\partial\epsilon}\bigg|_{\epsilon=0}, \qquad R^{(1)} = \frac{1}{\mathcal{F}^{\rm in}_{E,\infty}}\frac{\partial\mathcal{F}^{\rm ref}_{E,\infty}}{\partial\epsilon}\bigg|_{\epsilon=0}. \tag{50}$$

Thus the quantities plotted below are normalized changes of the horizon and infinity energy fluxes, rather than changes of radial-wave amplitudes alone. They are quoted for unit incident amplitude. For a general dimensionless incident amplitude $\mathcal{A}_{\rm in}$, homogeneity of the cubic source gives

$$\begin{aligned} T &= T^{(0)} + \epsilon|\mathcal{A}_{\rm in}|^2 T^{(1)} + \mathcal{O}(\epsilon^2|\mathcal{A}_{\rm in}|^4),\\ R &= R^{(0)} + \epsilon|\mathcal{A}_{\rm in}|^2 R^{(1)} + \mathcal{O}(\epsilon^2|\mathcal{A}_{\rm in}|^4). \end{aligned} \tag{51}$$

Accordingly, $T^{(1)}$ and $R^{(1)}$ are response coefficients per unit incident-amplitude squared; unlike their linear counterparts, they are not independent of the incident intensity. The current conservation identity gives the first-order balance law

$$T^{(1)} + R^{(1)} = 0. \tag{52}$$

The local reason for this identity is visible directly in the radial equation. Let $J^{(1)}$ be the term linear in ϵ in the radial Klein–Gordon current. The cross-current identity gives

$$\frac{{\rm d}J^{(1)}}{{\rm d}r} = -\,{\rm Im}\Big[\big(R^{\rm in}_{\ell m\omega}\big)^*\,Q_{\ell\ell m}(r)\Big] = 0, \tag{53}$$

because the self-channel projectors are real for a real spheroidal parameter and $Q_{\ell\ell m} \propto |R^{\rm in}|^2 R^{\rm in}$. Thus the smooth cubic source changes the two boundary currents through interference, but does not inject net Killing energy into the stationary mode at this perturbative order. Equation (52) is the boundary form of this local identity, while the field residual and spectral tail remain independent discretization diagnostics. Numerically this equation is the most useful global precision diagnostic, because it tests the homogeneous normalization, the nonlinear Green integral, and the flux normalization simultaneously. The fixed-incident normalization and the appearance of the $|B^{\rm inc}|^4$ denominators are derived explicitly in Appendix A.

2. *Numerical accuracy diagnostics*

The accuracy diagnostics used below are internal to the spectral calculation. Let the homogeneous equation

be written at a collocation point as

$$\mathcal{R}_R = \Delta R'' + \Delta' R' + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right) R. \tag{54}$$

The reported relative residual is

$$\delta_R = \frac{|\mathcal{R}_R|}{|\Delta R''| + |\Delta' R'| + |(K^2/\Delta - \lambda_{\ell m\omega})\, R|}. \tag{55}$$

Thus the denominator contains the actual local field amplitude, not only the differential-equation coefficients. This makes the residual insensitive to an overall rescaling of the homogeneous solution. The original radial residual is evaluated independently on 600 Chebyshev–Gauss points per subdomain that are not used in the collocation solve. The phase-peeled Chebyshev series is differentiated analytically at these off-grid points before reconstructing the three terms in Eq. (55); the degenerate endpoint rows remain reserved for the boundary regularity conditions. We also monitor the Chebyshev tail

$$\eta_{\rm tail} = \frac{\max_{N-15\le k\le N} |c_k|}{\max_{0\le k\le N} |c_k|}, \tag{56}$$

where c_k are the Chebyshev coefficients of the phase-peeled unknown on each subdomain. Finally, Eqs. (25) and (52) test the physical flux balance after the homogeneous matching and Green integral have both been performed.

The production data below retain the self channel $\ell' = \ell$ when quoting the first-order scattering coefficients. This is not an angular truncation of the first-order field: the cubic Kerr source in Eq. (29) does generate off-diagonal target channels. However, at order ϵ the total energy flux contains only interference between $\Phi^{(0)}$ and $\Phi^{(1)}$. Orthogonality of the spheroidal harmonics then removes every target channel except $\ell' = \ell$ from the flux correction. Off-diagonal channels are therefore required for the full first-order waveform, but not for the self-channel coefficients $T_\ell^{(1)}$ and $R_\ell^{(1)}$ reported here.

3. *Low-frequency behavior*

The scalar monopole provides an analytic low-frequency normalization check directly in the Kerr Teukolsky variable. Fix $|a| < M$, take $(\ell, m) = (0, 0)$, and introduce the far-zone coordinate $x = \omega r$. In the limit $\omega \to 0$ the normalized spheroidal harmonic and the self-channel projector become

$$S_{00} \longrightarrow \frac{1}{\sqrt{4\pi}}, \qquad C^{(0)}_{000} \longrightarrow \frac{1}{4\pi}. \tag{57}$$

The leading far-zone radial equation is the flat-space monopole equation. Matching to the solution that is unity at the horizon gives

$$\begin{aligned} R^{\rm in}_{00\omega} &= j_0(x) + o(1) = \frac{\sin x}{x} + o(1), \\ R^{\rm up}_{00\omega} &= \frac{e^{ix}}{r}[1 + o(1)]. \end{aligned} \tag{58}$$

Comparison with Eq. (22) and evaluation of Eq. (40) at infinity then yield

$$B^{\rm inc}_{00\omega} = \frac{\mathrm{i}}{2\omega}[1 + o(1)], \qquad W_0 = 2\mathrm{i}\omega B^{\rm inc}_{00\omega} = -1 + o(1). \tag{59}$$

The $a^2 C^{(2)}$ part of the source and all strong-field corrections are subleading relative to the far-zone $r^2 C^{(0)}$ term. Substitution of Eqs. (57)–(59) into the horizon Green integral therefore gives the full complex amplitude

$$\begin{aligned} A^{(1)}_{H,00} &= \frac{C^{(0)}_{000}}{\omega^2} \int_0^\infty \frac{\sin^3 x\, e^{ix}}{x^2}\, \mathrm{d}x + o(\omega^{-2}) \\ &= \frac{C^{(0)}_{000}}{\omega^2}(I_0 + \mathrm{i}J_0) + o(\omega^{-2}), \end{aligned} \tag{60}$$

where both integrals can be evaluated without a regulator. Using $\sin^3 x \cos x = \frac{1}{4}\sin 2x - \frac{1}{8}\sin 4x$ and integrating by parts,

$$\begin{aligned} I_0 &= \int_0^\infty \frac{\sin^3 x \cos x}{x^2}\, \mathrm{d}x \\ &= \frac{1}{2}\int_0^\infty \frac{\cos 2x - \cos 4x}{x}\, \mathrm{d}x = \frac{1}{2}\ln 2. \end{aligned} \tag{61}$$

while a second integration by parts gives

$$J_0 = \int_0^\infty \frac{\sin^4 x}{x^2}\, \mathrm{d}x = 4\int_0^\infty \frac{\sin^3 x \cos x}{x}\, \mathrm{d}x = \frac{\pi}{4}. \tag{62}$$

Since $C^{(0)}_{000} = 1/(4\pi)$, the amplitude and relative response are

$$\begin{aligned} A^{(1)}_{H,00} &= \left(\frac{\ln 2}{8\pi} + \frac{\mathrm{i}}{16}\right)\omega^{-2} + o(\omega^{-2}), \\ \rho_{H,00} &\longrightarrow \frac{\ln 2}{2\pi} + \frac{\mathrm{i}}{4}. \end{aligned} \tag{63}$$

The imaginary term is a finite horizon phase response after incident normalization; it does not contribute to the first-order energy flux. The real part and Eq. (47) give

$$\boxed{\lim_{\omega\to 0} \chi_{H,00} = 8C^{(0)}_{000} I_0 = \frac{\ln 2}{\pi} = 0.220635600153.} \tag{64}$$

The leading result is independent of spin because the limit is controlled by the far zone. The closed form $\ln 2/\pi$ is specific to the scalar monopole and is not a superradiant-threshold result: $m = 0$ implies $p = \omega$.

The same matching argument determines both absolute flux coefficients for every cubic axisymmetric self channel. Let $d = (M^2 - a^2)^{1/2}$ and first set $\omega = 0$. The horizon-regular radial equation and its unit-horizon solution are

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta \frac{\mathrm{d}R}{\mathrm{d}r}\right) - \ell(\ell+1)R = 0, \qquad R = P_\ell\left(\frac{r-M}{d}\right), \tag{65}$$

because $P_\ell(1)=1$. Since the leading coefficient of $P_\ell(z)$ is $q_\ell = (2\ell)!/[2^\ell(\ell!)^2]$, matching the growing part of Eq. (65) to the small-x limit of $j_\ell(x)$ gives

$$R^{\rm in}_{\ell0\omega} = \mathcal{K}_\ell (M\omega)^{-\ell} j_\ell(x)[1+o(1)],$$
$$\mathcal{K}_\ell = (2\ell+1)!!\, q_\ell \left(\frac{M}{d}\right)^\ell, \qquad x=\omega r. \tag{66}$$

The incoming part of j_ℓ fixes both the phase and magnitude,

$$B^{\rm inc}_{\ell0\omega} = \frac{M\mathrm{i}^{\ell+1}\mathcal{K}_\ell}{2(M\omega)^{\ell+1}}[1+o(1)],$$
$$|B^{\rm inc}_{\ell0\omega}|^2 = \frac{M^2\mathcal{K}_\ell^2}{4(M\omega)^{2\ell+2}}[1+o(1)]. \tag{67}$$

Substitution into Eq. (23), with $p=\omega$, yields the absolute linear law

$$T^{(0)}_{\ell0} = \mathcal{A}^{(0)}_\ell (M\omega)^{2\ell+2}[1+o(1)],$$
$$\mathcal{A}^{(0)}_\ell = \frac{4(r_+^2+a^2)}{M^2\mathcal{K}_\ell^2}. \tag{68}$$

For the nonlinear coefficient, define the spherical angular overlap and two convergent Bessel integrals

$$\mathcal{C}_\ell = \int |Y_{\ell0}|^4 \mathrm{d}\Omega, \qquad \mathcal{I}_\ell = -\int_0^\infty x^2 y_\ell(x) j_\ell^3(x)\mathrm{d}x,$$
$$\mathcal{J}_\ell = \int_0^\infty x^2 j_\ell^4(x)\mathrm{d}x > 0. \tag{69}$$

For fixed subextremal spin the $a^2C^{(2)}$ source term and the strong-field part of the Green integral are subleading. In units $M=1$, the outgoing solution and Wronskian in the far zone are $R^{\rm up} = \omega \mathrm{i}^{\ell+1} h^{(1)}_\ell(x)[1+o(1)]$ and $W_\ell = \mathrm{i}^{\ell+2}\mathcal{K}_\ell\omega^{-\ell}[1+o(1)]$. Their phases are essential. Since $\mathrm{i}h^{(1)}_\ell = \mathrm{i}j_\ell - y_\ell$, direct substitution in the Green integral gives, after restoring M,

$$\frac{A^{(1)}_{H,\ell0}}{M^2} = \mathcal{C}_\ell\mathcal{K}_\ell^2(\mathcal{I}_\ell + \mathrm{i}\mathcal{J}_\ell)(M\omega)^{-2\ell-2}[1+o(1)]. \tag{70}$$

Thus the unit-horizon coefficient has $|A^{(1)}_H| \propto (M\omega)^{-2\ell-2}$ and $\arg A^{(1)}_H \to \arg(\mathcal{I}_\ell + \mathrm{i}\mathcal{J}_\ell)$. This divergence is not a divergent scattering response: maintaining unit horizon amplitude requires a growing incident wave as $\omega\to0$. Combining Eqs. (67) and (70) instead gives

$$MA^{(1)}_{H,\rm fix} = \frac{8\mathcal{C}_\ell}{\mathcal{K}_\ell}\mathrm{i}^{-(\ell+1)}(\mathcal{I}_\ell+\mathrm{i}\mathcal{J}_\ell)(M\omega)^{\ell+1}[1+o(1)],$$
$$\rho_{H,\ell0} \longrightarrow \rho^{\rm IR}_{H,\ell} = 4\mathcal{C}_\ell(\mathcal{I}_\ell+\mathrm{i}\mathcal{J}_\ell). \tag{71}$$

The fixed-incident first-order amplitude therefore vanishes as $(M\omega)^{\ell+1}$, while its ratio to the linear horizon amplitude tends to a finite complex constant. Taking its real part cancels the near-zone matching coefficient and recovers the flux laws

$$\chi_{H,\ell0} \longrightarrow \chi^{\rm IR}_\ell = 2\,\mathrm{Re}\,\rho^{\rm IR}_{H,\ell} = 8\mathcal{C}_\ell\mathcal{I}_\ell,$$
$$T^{(1)}_{\ell0} = \mathcal{A}^{(1)}_\ell (M\omega)^{2\ell+2}[1+o(1)], \qquad \mathcal{A}^{(1)}_\ell = \mathcal{A}^{(0)}_\ell \chi^{\rm IR}_\ell. \tag{72}$$

Thus the cubic interaction is infrared marginal for every axisymmetric self channel: it changes the coefficient, but not the linear absorption power. The ratio $\chi^{\rm IR}_\ell$ is independent of spin, although the absolute coefficients retain spin dependence through $\mathcal{K}_\ell$ and the horizon-area factor. This statement takes $\omega\to0$ at fixed $|a|<M$; Eq. (65) also makes clear why the limit is not uniform as $a/M\to1$ for $\ell>0$.

Table I evaluates the new formulas for the $a/M=0.5$ production spin. The Bessel integrals were summed over successive π intervals and extrapolated in inverse endpoint powers. Independent Kerr solves at $(N,q)=(144,144)$ and $(176,176)$ used only the horizon-side response at the listed lowest frequencies. At the higher resolution the relative differences between the scaled numerical and analytic coefficients are $(0.48, 0.96, 3.32)\%$ for $T^{(0)}$ and $(0.82, 1.07, 3.41)\%$ for $T^{(1)}$, ordered by $\ell=0,1,2$; these are finite-frequency corrections, not fitted exponents. The two-resolution changes in χ_H are given in the final column, and the largest Wronskian error is 2.5×10^{-11}. The same runs test information absent from the energy flux. For $\ell=0,1,2$ they give $\mathcal{J}_\ell = (0.785398163397, 0.366519142919, 0.246839422782)$ and limiting phases $(66.1895^\circ, 70.6933^\circ, 72.4699^\circ)$. At the lowest listed frequencies the complex relative responses agree with Eq. (71) to $(0.228, 0.148, 0.318)\%$, while their simultaneous-resolution changes are $(6.3\times10^{-6}, 5.0\times10^{-5}, 3.0\times10^{-4})$. Hence both the dissipative real part and the reactive phase part are independently resolved. The infinity-side $R^{(1)}$ interference term is already cancellation limited at these frequencies and is deliberately not used in this infrared audit.

As a numerical test of the powers rather than only their coefficients, we fit the seven horizon-side points for each multipole freely in log–log space. For $T^{(0)}$ the fitted powers are 2.0145, 4.0298, and 6.0784; for $T^{(1)}$ they are 2.0245, 4.0360, and 6.0884. Every fit has $R^2>0.999989$. Restricting to the lowest five frequencies moves the two sets to $(2.0095, 4.0193, 6.0502)$ and $(2.0162, 4.0237, 6.0585)$, respectively, showing convergence toward the exact powers $(2,4,6)$ as the upper end of the fit window is removed. The analytic conclusion still comes from matched asymptotics; these free fits are an independent finite-frequency audit. For $\ell=2$, attempts at $M\omega=0.003$ and 0.004 failed the paired continuation-normalization check and are excluded, so the reported infrared window begins at $M\omega=0.005$.

This calculation extends analytically to every positive integer n. Consider the $U(1)$-invariant local equation

$$\Box\Phi + \epsilon_n|\Phi|^{2n}\Phi = 0, \qquad n=1,2,\ldots, \tag{73}$$

TABLE I. Analytic cubic infrared coefficients and direct horizon-side checks for $a/M = 0.5$. Here $T^{(i)}_{\ell 0} \sim \mathcal{A}^{(i)}_\ell (M\omega)^{2\ell+2}$ and $\delta_N \chi_H$ compares the simultaneous $(N, q) = (144, 144)$ and $(176, 176)$ calculations at the lowest listed frequency.

ℓ	$\mathcal{I}_\ell$	$\chi^{\rm IR}_\ell$	$\mathcal{A}^{(0)}_\ell$	$\mathcal{A}^{(1)}_\ell$	lowest $M\omega$	$\delta_N \chi_H$
0	0.346573590280	0.220635600153	14.9282032	3.29369308	0.00075	1.1×10^{-7}
1	0.128401008797	0.147136717786	1.24401694	0.183040569	0.00150	1.5×10^{-4}
2	0.077970747421	0.106366541731	0.0165868925	0.00176429039	0.00500	7.7×10^{-5}

for which $[\epsilon_n] = M^{2n-2}$ when $[\Phi] = M^{-1}$. The cubic model used elsewhere in this paper is the $n = 1$ member. At first order the source scales as $|\alpha|^{2n}\alpha$ under $R^{(0)} \mapsto \alpha R^{(0)}$, so fixed-incident normalization gives

$$T^{(1)}_n = \frac{2p(r_+^2 + a^2)}{\omega |B^{\rm inc}|^{2n+2}} \operatorname{Re} A^{(1)}_{H,n},$$
$$\chi^{(n)}_H \equiv \frac{T^{(1)}_n}{T^{(0)}} = \frac{2 \operatorname{Re} A^{(1)}_{H,n}}{|B^{\rm inc}|^{2n}}. \tag{74}$$

For a general incident mode and target multipole the projected source is

$$Q^{(n)}_{\ell'\ell m}(r) = \left[r^2 C^{(0,n)}_{\ell'\ell m} + a^2 C^{(2,n)}_{\ell'\ell m} \right] |R^{\rm in}_{\ell m\omega}|^{2n} R^{\rm in}_{\ell m\omega}, \tag{75}$$
$$C^{(j,n)}_{\ell'\ell m} = \int \mathrm{d}\Omega \, \cos^j\theta \, S^*_{\ell' m} |S_{\ell m}|^{2n} S_{\ell m}, \qquad j = 0, 2. \tag{76}$$

The Green amplitudes follow from the same in/up Wronskian as in the cubic calculation, with $Q \mapsto Q^{(n)}$. In particular,

$$A^{(1)}_{H,n} = -\frac{1}{W_\ell} \int_{r_+}^\infty Q^{(n)}_{\ell\ell m}(r) R^{\rm up}_{\ell m\omega}(r) \mathrm{d}r. \tag{77}$$

For a dimensionless incident amplitude $\mathcal{A}_{\rm in}$, the self-channel flux expansion and its conservation law are

$$T = T^{(0)} + \epsilon_n |\mathcal{A}_{\rm in}|^{2n} T^{(1)}_n + \mathcal{O}(\epsilon_n^2 |\mathcal{A}_{\rm in}|^{4n}),$$
$$R = R^{(0)} + \epsilon_n |\mathcal{A}_{\rm in}|^{2n} R^{(1)}_n + \mathcal{O}(\epsilon_n^2 |\mathcal{A}_{\rm in}|^{4n}),$$
$$T^{(1)}_n + R^{(1)}_n = 0. \tag{78}$$

This makes $M^{2n-2}\chi^{(n)}_H$ dimensionless. Numerically, the outer integrand is not formed by raising a cancellation-prone oscillatory sum to a large power. We expand $(B^{\rm inc} e^{-i\omega r_*} + B^{\rm ref} e^{i\omega r_*})^{n+1}$ and its conjugate factor binomially, multiply by the target homogeneous wave, and integrate the resulting finite phase channels separately. This gives $2n + 3$ channels with phase indices $q = -2n - 2, -2n, \ldots, 2n + 2$ and asymptotic envelope r^{-2n}. The construction is therefore finite for every fixed positive integer n and retains oscillatory-tail control as the operator order is increased.

For the monopole the angular projections reduce to

$$C^{(0)}_n = \int |S_{00}|^{2n+2} \mathrm{d}\Omega = (4\pi)^{-n}, \qquad C^{(2)}_n = \frac{1}{3} C^{(0)}_n. \tag{79}$$

The leading far-zone Green integral is consequently

$$\operatorname{Re} A^{(1)}_{H,n} = \frac{C^{(0)}_n}{\omega^2} I_n + o(\omega^{-2}),$$
$$I_n = \int_0^\infty \frac{\sin^{2n+1} x \cos x}{x^{2n}} \mathrm{d}x,$$
$$= \frac{n}{n+1} \int_0^\infty \frac{\sin^{2n+2} x}{x^{2n+1}} \mathrm{d}x > 0. \tag{80}$$

The second form follows by integration by parts and proves positivity without numerics. Repeated integration by parts, followed by the standard finite cosine expansion of $\sin^{2n+2} x$, gives the regulator-independent finite sum

$$I_n = \frac{(-1)^n}{4(n+1)(2n-1)!} \sum_{j=1}^{n+1} (-1)^{j+1} j^{2n} \binom{2n+2}{n+1-j} \ln j. \tag{81}$$

Using $B^{\rm inc} = \mathrm{i}/(2\omega) + o(\omega^{-1})$ in Eq. (74) yields the all-n infrared law

$$M^{2n-2}\chi^{(n)}_{H,00} = c_n (M\omega)^{2n-2} + o\big((M\omega)^{2n-2}\big), \tag{82}$$
$$c_n = \frac{2I_n}{\pi^n} > 0.$$

Thus $n = 1$ is infrared marginal and reproduces $c_1 = \ln 2/\pi$, whereas every $n > 1$ operator is infrared irrelevant in this scattering observable. The next two coefficients are

$$c_2 = 0.0631722552062, \qquad c_3 = 0.0172436384309. \tag{83}$$

For the full $\ell = 0, 1, 2$ spectra, the far-zone normalization and the covariant source weight $r^2 C^{(0)}_{\ell\ell m} + a^2 C^{(2)}_{\ell\ell m}$ still control how quickly a finite-frequency calculation reaches Eqs. (68) and (72). We therefore retain a separate globally balanced Kerr experiment for $a = 0.5$ and $m = 0$, rather than use the asymptotic formulas to manufacture display points.

The infinity-side interference integral becomes cancellation limited before the transmitted coefficient loses smoothness. For points that test both boundary fluxes, we therefore apply the same global requirement used throughout the paper,

$$|\delta_1| = |T^{(1)} + R^{(1)}| \le 10^{-8}, \tag{84}$$

before a point can enter the globally balanced part of a figure or a two-boundary physical fit. Table II gives the

resulting displayed windows. The lowest retained frequencies are $M\omega = 0.02$, 0.12, and 0.28 for $\ell = 0, 1, 2$, respectively. The filled data in Fig. 2 show that both $T^{(0)}$ and $T^{(1)}$ decrease rapidly toward the low-frequency side. The additional open points are the paired horizon-side calculations of Table I; they extend below the reflected-channel cancellation boundary and are compared with the analytic leading laws. We do not fit a power law to the globally balanced segments. The free infrared fits quoted above use only the open horizon-side points, while the powers in Eqs. (68) and (72) come from matched asymptotics.

We test each lowest retained point independently. Starting from a $N = 256$, $q = 256$, $r_{\rm m} = 40M$ reference, we lower N to 224, lower q to 192, and move the matching radius to $32M$ and $48M$. Table III shows that the largest relative change in $T^{(1)}$ is 1.8×10^{-8}; the largest change in $R^{(1)}$ is 2.0×10^{-8}, and every run satisfies the first-order balance at or below 3.1×10^{-12}. The displayed low-frequency-side trend is therefore supported by both boundary fluxes, rather than inferred from a cancellation-limited reflected amplitude.

We next test the all-n prediction in Eq. (82) without using it as an input to the radial solve. For $n = 2, 3$ we computed the $\ell = m = 0$ response at $M\omega = (0.0005, 0.00075, 0.001, 0.0015, 0.002, 0.003, 0.005)$ and $a/M = (0, 0.5, 0.9, 0.99, 0.999)$, independently at $(N, q) = (144, 144)$ and $(176, 176)$. A quadratic extrapolation of $\chi_H^{(n)}/\omega^{2n-2}$ over the lowest five frequencies recovers the analytic coefficient at every spin. Changing the fit to the lowest four or six points changes the extrapolated coefficient by at most 4.6×10^{-6}, and the spin-to-spin relative standard deviations of the intercepts are 1.2×10^{-6} and 7.2×10^{-7} for $n = 2, 3$. The direct log–log slopes in the finite fit window are slightly above their asymptotic integer values, with the ranges given in Table IV; their R^2 values exceed 0.9999989.

The fixed-frequency limit does not by itself test whether the conclusion is uniform as extremality is approached. We therefore performed a second scan at fixed $y = \omega/\kappa = (0.025, 0.05, 0.1)$ for $a/M = (0.98, 0.99, 0.995, 0.997, 0.999)$. Equation (82) predicts

$$M^{2n-2}\chi_{H,00}^{(n)} \sim c_n y^{2n-2}(M\kappa)^{2n-2}. \tag{85}$$

At the smallest y, fits to the three highest spins give powers 2.00734 and 4.01128, with $R^2 = 0.99999988$ and 0.99999993. The results therefore resolve suppression by κ^2 and κ^4, respectively. This is qualitatively different from the $1/\kappa$ growth of the cubic principal $\ell = m \geq 2$ channels: increasing the local power does not manufacture a monopole throat enhancement.

The new source calculations reuse the same homogeneous solution for every n. At the lowest representative point $(a/M, M\omega, N) = (0.5, 5 \times 10^{-4}, 176)$, the last 16 normalized phase-peeled Chebyshev coefficients are at most 5.2×10^{-14}. Across 1800 independent Chebyshev–Gauss validation points, the original field-dependent radial residual in Eq. (55) has median 2.7×10^{-10}, 95th percentile 2.1×10^{-7}, and a localized maximum 1.6×10^{-5}. Over both response scans the largest two-resolution change is 2.5×10^{-6}, the largest linear-balance error is 7.7×10^{-13}, and the largest Wronskian error is 1.1×10^{-14}. At the smallest responses the separately reconstructed reflected interference is cancellation limited and is not used: the power-law test uses the direct horizon Green integral, its two-resolution convergence, and the independent homogeneous diagnostics just quoted.

OpenAI Codex (GPT-5, accessed August 2026) assisted with implementation and debugging of the generalized source, construction of the parameter scans, and numerical postprocessing. The authors specified the physical problem and acceptance tests, executed the calculations locally, and independently checked the analytic limits, paired-resolution results, conservation laws, Wronskians, and manuscript tables against the frozen data files.

4. *High-frequency behavior*

At high frequencies the phase-peeled homogeneous amplitudes become slowly varying, while the nonlinear Green integral is controlled by rapidly oscillatory tails. Motivated by the thermal scale associated with a black-hole horizon [60] and by finite-window Boltzmann diagnostics in the Kerr graybody-factor literature [15, 16], we test an exponential finite-window diagnostic. We do not infer this form from a nonrotating asymptotic integral, because the Teukolsky radial normalization and the Σ-weighted source modify the integrand. The model is

$$T_\ell^{(1)}(\omega) \simeq A_\ell \exp\left[-\frac{\omega}{T_{\rm fit}}\right], \qquad \omega > {\rm Re}\,\omega_{\ell 0}. \tag{86}$$

For Kerr

$$T_H = \frac{r_+ - r_-}{4\pi(r_+^2 + a^2)}. \tag{87}$$

For $a = 0.5$ the fits in Sec. IV give $\pi M T_{\rm fit} = 0.111$, 0.105, and 0.101 for $\ell = 0, 1, 2$, close to $\pi M T_H = 0.1160$. These values should be read as finite-window effective temperatures rather than as a proof of an exact high-frequency theorem.

IV. RESULTS

Our main result on the Kerr nonlinear scattering coefficient is shown in Fig. 3. It plots $T_\ell^{(1)}$ for $\ell = 0, 1, 2$, uses a linear scale to display the resonance peaks, and uses a logarithmic scale to display the high-frequency tail. The dashed lines mark the real parts of the fundamental scalar Kerr quasinormal modes computed with

TABLE II. Accepted low-frequency-side data displayed in Fig. 2. Only points satisfying $|T^{(1)} + R^{(1)}| \leq 10^{-8}$ are retained. The last three columns give the linear transmission, first-order transmission, and absolute first-order balance residual at the lowest displayed frequency.

ℓ	displayed $M\omega$ range	points	$T^{(0)}_{\rm low}$	$T^{(1)}_{\rm low}$	$\lvert\delta_1\rvert_{\rm low}$
0	$[0.02, 0.09]$	15	6.92348×10^{-3}	1.67885×10^{-3}	1.4×10^{-13}
1	$[0.12, 0.28]$	17	9.73095×10^{-4}	1.78519×10^{-4}	3.7×10^{-14}
2	$[0.28, 0.46]$	19	4.35440×10^{-4}	6.58814×10^{-5}	2.0×10^{-14}

TABLE III. Convergence audit at the lowest accepted displayed point for each multipole. The reference uses $(N, q, r_{\rm m}/M) = (256, 256, 40)$. The maxima include the active $(240, 192, 40)$ setting and the one-choice variations $(224, 256, 40)$, $(256, 192, 40)$, $(256, 256, 32)$, and $(256, 256, 48)$. The last column is the largest absolute first-order balance residual in the scan.

ℓ	$M\omega$	$T^{(1)}_{\rm ref}$	$\max\lvert\Delta T^{(1)}/T^{(1)}\rvert$	$\max\lvert\Delta R^{(1)}/R^{(1)}\rvert$	$\max\lvert\delta_1\rvert$
0	0.02	1.67885×10^{-3}	1.8×10^{-8}	2.0×10^{-8}	3.0×10^{-12}
1	0.12	1.78519×10^{-4}	6.7×10^{-9}	7.3×10^{-9}	3.8×10^{-13}
2	0.28	6.58814×10^{-5}	9.2×10^{-9}	4.6×10^{-9}	2.8×10^{-13}

the local `qnm` package [61] for $s = 0$, $a = 0.5$, $m = 0$, and $n = 0$.

Three features are robust in the accepted data. First, in the axisymmetric Kerr cases studied here, $T^{(1)}_\ell$ is positive at every accepted sampled frequency in the present flux convention. For positive coupling in Eq. (5), the nonlinear correction therefore increases the absorbed flux and, by Eq. (52), decreases the reflected flux at the same order. This sign statement refers to the coefficient convention of Eqs. (45) and (46); it should not be compared to a reduced Schwarzschild field convention without translating the flux normalization.

Second, the peaks of $T^{(1)}_\ell$ lie on the frequency scale set by the real parts of the corresponding fundamental Kerr scalar QNMs [8, 61]. The fitted peak positions are $M\omega_{\rm peak} = 0.1306$, 0.3101, and 0.5026 for $\ell = 0, 1, 2$, while the QNM real parts are 0.1124, 0.2979, and 0.4920. The corresponding fractional offsets are 16.2%, 4.1%, and 2.2%. To make the comparison quantitative without identifying the flux with a pole, we fit the selected samples to the four-parameter profile

$$T^{(1)}_\ell(\omega) = T_{\rm off} + \frac{A\,\gamma^2}{(M\omega - \omega_p)^2 + \gamma^2}. \tag{88}$$

Here ω_p is the fitted center in units $M = 1$, and γ is the half-width parameter of the phenomenological Breit–Wigner-type interpolation, not a claim that the flux itself is a meromorphic response. The coefficient of determination of the three fits is $R^2 = 0.9918$, 0.9978, and 0.9985 for $\ell = 0, 1, 2$, respectively. The comparison is qualitative rather than a precision QNM determination: the nonlinear coefficient is a real-frequency flux observable, whereas the QNM frequencies are poles of the analytically continued response. A genuine Kerr QNM projection requires a conserved bilinear form and contour regularization [38], neither of which is part of this fit. The fit uses the sampled points satisfying $T^{(1)}_\ell > 0.2T^{(1)}_{\ell,\rm max}$; the relative RMS residuals of these peak fits over the selected points are 2.41%, 1.27%, and 1.05% for $\ell = 0, 1, 2$, respectively. These residuals quantify the descriptive quality of the interpolation, not an error bar on the QNM frequencies. As a fit-window sensitivity check, we repeat the selection with thresholds from 0.15 to 0.30 of the sampled maximum, retaining at least five points in every fit. The resulting fitted centers vary by at most 6.7×10^{-4}, 3.9×10^{-4}, and 3.4×10^{-4} for $\ell = 0, 1, 2$, respectively, whereas the corresponding half-widths vary by 3.1×10^{-3}, 4.8×10^{-3}, and 5.2×10^{-3}. The center locations are therefore stable at the resolution of the sweep, while the widths should be regarded as descriptive fit parameters. The complete threshold scan is included in the reproducibility data.

Third, at high frequency the accepted tail points are well fit by the Boltzmann form Eq. (86). The fitted temperatures differ from the Kerr Hawking temperature by 4.0%, 9.8%, and 12.8% for $\ell = 0, 1, 2$, respectively. These are effective finite-window fit parameters, not an analytic derivation of a thermal spectrum. On the accepted low-frequency side the Kerr fixed-incident coefficients decrease as ω is lowered over the globally balanced windows. The deeper horizon-side data then recover the matched Kerr powers $(M\omega)^{2\ell+2}$ with free-fit $R^2 > 0.999989$. This differs from simply copying the Schwarzschild reduced-field statement and is therefore reported as a Kerr analytic result with an independent numerical audit.

The exponential fits use unweighted least squares for $\log T^{(1)}_\ell$ versus $M\omega$ over the windows $[0.18, 0.60]$, $[0.38, 0.85]$, and $[0.62, 1.00]$ for $\ell = 0, 1, 2$, respectively. The corresponding root-mean-square residuals in log space are 3.7×10^{-2}, 6.7×10^{-2}, and 5.6×10^{-2}. These window-dependent residuals quantify the quality of the finite-window fit; they are not estimates of a systematic error outside the computed frequency range.

TABLE IV. Scalar-monopole test of the $|\Phi|^{2n}\Phi$ infrared hierarchy. The fixed-ω slope is the range over the five spins; the fixed-y slope uses $y = 0.025$ and the three highest spins. The coefficient error is the largest quadratic-extrapolation error over spin. The final column is the largest two-resolution response change over both scans.

n	c_n	predicted power	fixed-ω fitted power	fixed-y κ power	max. coefficient error	max. $\delta_{N\chi_H}$
2	0.0631722552	2	2.01009–2.01016	2.00734	2.8×10^{-6}	2.5×10^{-6}
3	0.0172436384	4	4.01552–4.01561	4.01128	1.3×10^{-6}	7.7×10^{-9}

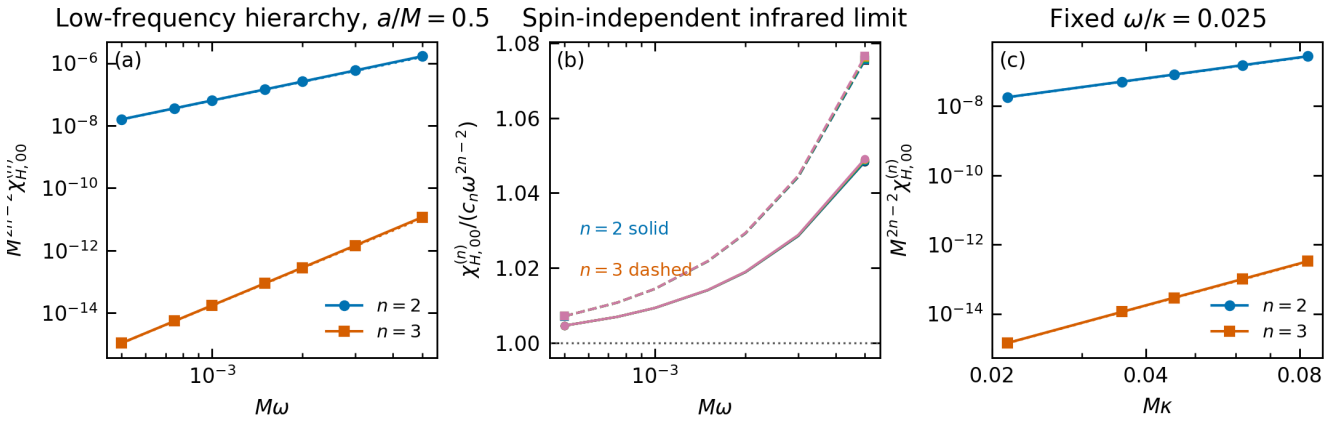


FIG. 1. Higher-power scalar-monopole response. (a) Direct horizon response for $n = 2, 3$ at $a/M = 0.5$; dotted lines are the analytic leading laws $c_n\omega^{2n-2}$. (b) The scaled response divided by c_n for five spins; solid circles and dashed squares denote $n = 2$ and 3. The near coincidence of the colored spin curves exhibits the common far-zone limit. (c) At fixed $\omega/\kappa = 0.025$, the response decreases toward extremality; dotted curves are $c_n(0.025\kappa)^{2n-2}$. Points show the $N = q = 176$ calculation; all pass the paired-resolution, linear-balance, and Wronskian tests described in the text.

TABLE V. Fitting parameters of the Kerr nonlinear transmission resonance and the high-frequency Boltzmann tail. The QNM frequencies are scalar Kerr fundamental modes with $s = 0$, $a = 0.5$, $m = 0$, and $n = 0$. All frequencies are reported in units $M = 1$.

ℓ	ω_p	γ	$\Re\omega_Q$	$\lvert\Im\omega_Q\rvert$	$\pi T_{\rm fit}$	ΔT
0	0.1306	0.0599	0.1124	0.1022	0.1114	4.00%
1	0.3101	0.0591	0.2979	0.0954	0.1046	9.85%
2	0.5026	0.0595	0.4920	0.0946	0.1012	12.81%

Because the effective temperature is a fit diagnostic rather than an analytic observable, we also repeat the tail fit after shifting either edge of each window while retaining at least three positive data points. The resulting window ranges and the largest logarithmic fit residual are given in Table VI. The fitted values vary only at the percent level within these local windows, while their absolute offsets from T_H remain consistent with the finite-window qualification above.

To test whether a power law can mimic the exponential over these finite windows, we fit the same positive data to $\log T^{(1)} = c - \beta M\omega$ and $\log T^{(1)} = c - n\log(M\omega)$. Both models have two fitted parameters, so their corrected Akaike information criteria can be compared without a parameter-count ambiguity. For n_s samples and $k = 2$ fitted parameters we use

$$\mathrm{AICc} = n_s \log(\mathrm{RSS}/n_s) + 2k + \frac{2k(k+1)}{n_s - k - 1}, \qquad (89)$$

with the residual sum of squares evaluated in log space. Table VII also reports leave-one-out prediction residuals. The exponential has the smaller AICc and the smaller predictive residual for every ℓ. A fit with the exponential slope fixed exactly to $-1/T_H$ has logarithmic RMS residuals 0.139, 0.443, and 0.514 for $\ell = 0, 1, 2$; hence the data support a finite-window exponential more strongly than either a power law or an exact thermal slope. This model comparison strengthens the descriptive tail claim but does not turn it into an asymptotic theorem.

All peak and tail fits, and all filled sweep points in Figs. 2 and 3, use only the accepted globally balanced subset. The open infrared points in Fig. 2 instead obey the paired horizon-side and Wronskian checks stated above. The largest first-order balance residual of the global subset is 6.9×10^{-12}, 2.2×10^{-12}, and 1.5×10^{-12} for $\ell = 0, 1, 2$, respectively. Completed solver rows that fail the 10^{-8} gate remain in the reproducibility data as an audit trail, but they are omitted from the plots, tables, peak interpolation, and low-frequency discussion. No physical claim is based on those excluded rows.

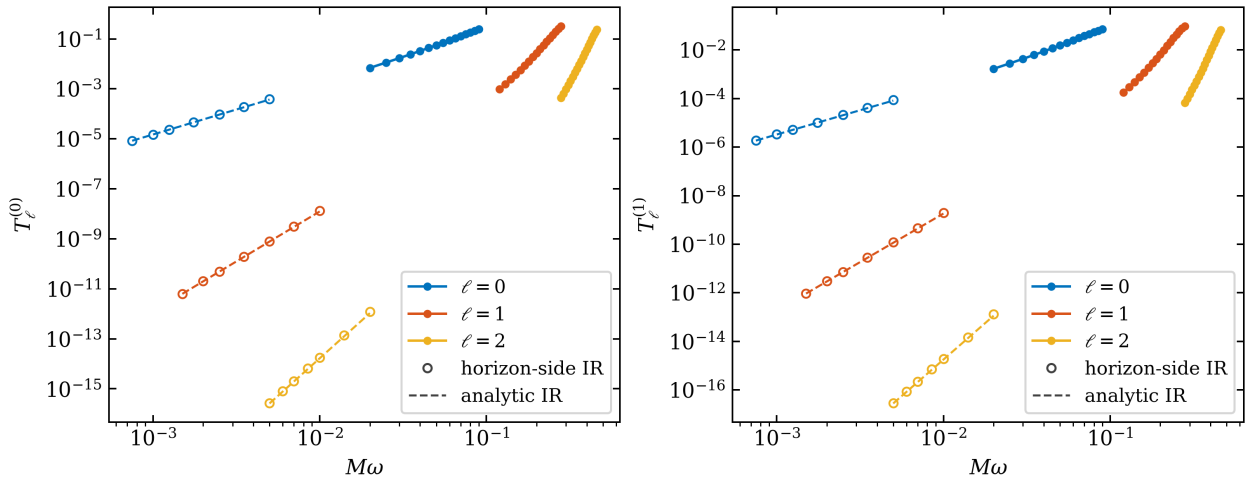


FIG. 2. Low-frequency Kerr diagnostics for $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$. Left: linear transmission coefficients. Right: self-channel nonlinear transmission coefficients. Filled circles and solid segments are the dense globally balanced sweep satisfying $|T^{(1)} + R^{(1)}| \leq 10^{-8}$. Open circles are the higher-resolution member of the paired horizon-side infrared audit; dashed segments are the analytic leading laws $\mathcal{A}_\ell^{(i)}(M\omega)^{2\ell+2}$. The gap between the two sets marks loss of reflected-channel cancellation accuracy, not loss of convergence of the transmitted coefficient.

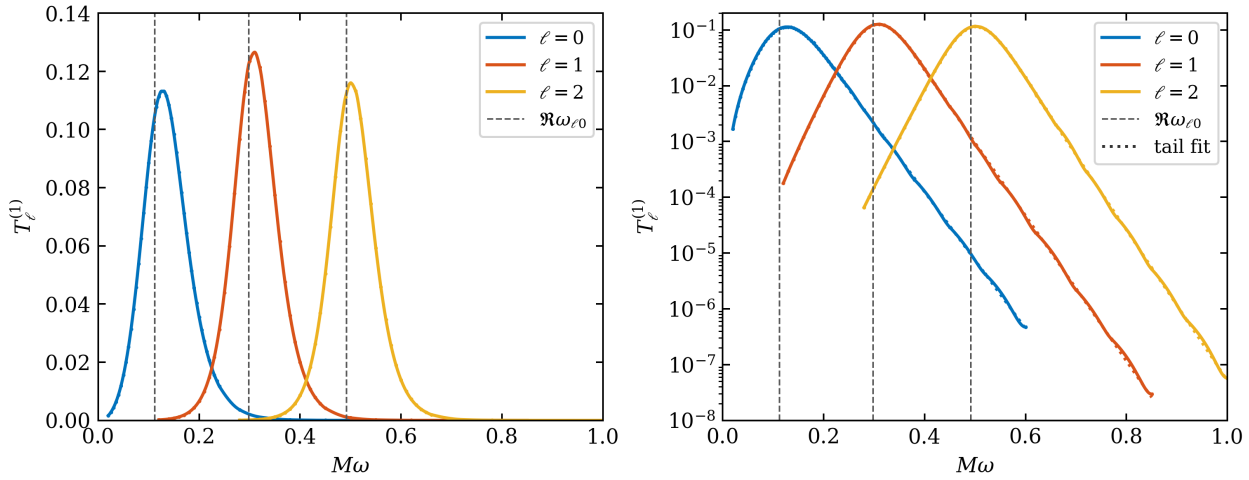


FIG. 3. Nonlinear Kerr transmission coefficients $T_\ell^{(1)}$ for $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$. The left panel uses a linear scale to show the Breit–Wigner-type peaks. The right panel uses a logarithmic scale to show the exponential high-frequency tails; dotted segments show the finite-window fits used in Table V. Dashed vertical lines mark the real parts of the fundamental scalar Kerr QNMs $\mathrm{Re}\,\omega_{\ell 0}$. All 43 points entering the three tail-fit windows use $q = 2048$; the denser low-frequency and peak samples use $q = 192$. Circles show every accepted sample, while the solid curves are shape-preserving interpolants used only as visual guides. Every plotted point satisfies $|T^{(1)} + R^{(1)}| \leq 10^{-8}$.

The change from the former production resolution to the active refined resolution is quantified separately in Table IX. At the representative peak points the refined calculation changes the transmission coefficient by at most 3.2×10^{-5} and preserves the first-order flux balance at the 10^{-12} level. We therefore use the refined curves for the physical discussion below; the lower-resolution values are reported to make the resolution choice auditable rather than folded into an unreported systematic error.

The high-frequency tail requires a separate quadrature audit because flux balance alone does not detect a common error in the two interference terms. At fixed $N = 240$, the compact inner integral was scanned from $q = 192$ to $q = 2048$, with twelve intermediate orders, at the three representative points in Table X. The sequence is mildly nonmonotonic at intermediate order but reaches a plateau at the two largest orders. We therefore recomputed every point entering the tail fits at $q = 2048$. Lowering N at fixed $q = 2048$ produces a much smaller change, showing that the remaining sensitivity belongs to the oscillatory Green integral rather than to the homogeneous spectral solution.

TABLE VI. Sensitivity of the finite-window effective temperatures to the fit window. The baseline window is the one used in Table V; the range is obtained from three nearby edge-shifted windows. The final column is the largest absolute fractional offset from the Kerr Hawking temperature among the four windows.

ℓ	baseline $[M\omega_{\min}, M\omega_{\max}]$	$\pi T_{\rm fit}^{\rm base}$	range of $\pi T_{\rm fit}$	$\max \sigma_{\log}$	$\max \lvert T_{\rm fit}/T_H - 1\rvert$
0	$[0.18, 0.60]$	0.1114	$[0.1109, 0.1122]$	3.86×10^{-2}	4.45%
1	$[0.38, 0.85]$	0.1046	$[0.1037, 0.1068]$	6.95×10^{-2}	10.61%
2	$[0.62, 1.00]$	0.1012	$[0.1002, 0.1031]$	5.59×10^{-2}	13.62%

TABLE VII. Finite-window comparison of exponential and power-law descriptions of the high-frequency nonlinear transmission tail. The residuals are RMS errors in $\log T^{(1)}$; ΔAICc is $\mathrm{AICc}_{\rm power} - \mathrm{AICc}_{\rm exp}$, so positive values favor the exponential. The final two columns are leave-one-out logarithmic prediction residuals.

ℓ	points	$\sigma_{\rm exp}$	$\sigma_{\rm power}$	ΔAICc	$\sigma_{\rm LOO,exp}$	$\sigma_{\rm LOO,power}$
0	19	3.72×10^{-2}	4.89×10^{-1}	97.9	4.56×10^{-2}	6.09×10^{-1}
1	14	6.71×10^{-2}	3.75×10^{-1}	48.2	8.92×10^{-2}	4.80×10^{-1}
2	10	5.59×10^{-2}	1.84×10^{-1}	23.8	8.11×10^{-2}	2.62×10^{-1}

We additionally test the self channel with $\ell = m = 2$ at both signs of the Kerr horizon frequency p and for a rapidly rotating case. Table XI records finite-resolution changes under N, $r_{\rm m}$, and q; these are not rigorous error bounds. The horizon amplitude is less sensitive than the reflected amplitude, while the Wronskian error remains below 6×10^{-16}. We therefore use flux balance, rather than one amplitude difference, as the acceptance criterion.

The angular discretization is inexpensive enough to audit independently of the radial Green calculation. Table XII tests both the largest spheroidal parameter in the displayed axisymmetric data, $(\ell, m, M\omega; \ell') = (2, 0, 1.0; 2)$, and the weakest nonzero channel displayed in Table XIII, $(2, 2, 0.30; 6)$. The active angular settings change the separation constants by at most 4.4×10^{-16} and the source projectors by at most 5.3×10^{-14} relative to the reference. Lowering the basis and quadrature separately gives changes of the same order or less, so angular discretization is not the limiting error in the reported data.

Local field diagnostic.
At the representative peak in Fig. 4, the 600-point Chebyshev–Gauss grid on each of the three homogeneous branches gives 1800 independent samples. Across the 1800 off-grid validation points, the field-dependent radial residual has maximum, 95th-percentile, and median values 2.8×10^{-10}, 1.2×10^{-11}, and 3.2×10^{-13}.
The largest normalized Chebyshev tail is 2.8×10^{-14} and the Wronskian mismatch is below 3×10^{-16}. These local diagnostics are reported separately from the global flux-balance residual.

A. Target-channel structure

The spheroidal source projection is a genuinely Kerr-specific part of the calculation. To expose it directly, we evaluated the target channels $\ell' = 2, \ldots, 6$ for the representative mode $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$ using $N = 288$, quadrature order $q = 224$, and Fourier-tail tolerance 10^{-11}. The amplitudes in Table XIII use the unit-horizon normalization of Eq. (21); they are not fixed-incident flux coefficients. Odd target channels vanish by the even-parity angular projectors in Eq. (29), while the even channels span several orders of magnitude in the outgoing Green amplitude.

The channel scan is converged for the dominant reflected amplitudes: changing N from 256 to 288 changes $A_{\rm ref}^{(1)}$ by 2.5×10^{-7} in the self channel and by 6.3×10^{-7} in the $\ell' = 4$ channel. The corresponding $\ell' = 4$ horizon amplitude changes by 7.1×10^{-3}, whereas the $\ell' = 6$ horizon amplitude is close to the double-precision cancellation floor and changes by order unity. For the displayed $\ell' = 6$ row, the independent diagnostic $(|B_{\rm inc}|^2 + |B_{\rm ref}|^2)/||B_{\rm inc}|^2 - |B_{\rm ref}|^2|$ is 3.27×10^{13}; this is the relevant conditioning indicator for the horizon-normalized continuation. We therefore use this experiment to establish the target-channel structure and the dominant outgoing amplitudes, not to claim uniform high precision for every off-diagonal horizon coefficient. Since off-diagonal channels do not interfere with the zeroth-order field, their energy flux first enters at $\mathcal{O}(\epsilon^2)$.

B. Superradiant threshold response and spin dependence

The axisymmetric production sweep removes the most distinctive rotating black-hole effect because $p = \omega$ when $m = 0$. We therefore extend the self channel to $a/M = 0.5, 0.7, 0.9$ and $\ell = m = 1, 2$, using the threshold coordinate

$$x = \frac{\omega}{m\Omega_H}. \tag{90}$$

TABLE VIII. Numerical quality of the axisymmetric Kerr frequency sweeps. Here $\delta_0 = |T^{(0)} + R^{(0)} - 1|$ and $\delta_1 = |T^{(1)} + R^{(1)}|$. Every retained point satisfies $\delta_1 \le 10^{-8}$. All active points use $N = 240$. The low-frequency and peak samples use $q = 192$. All 43 samples in the three tail-fit windows use $q = 2048$ after the audit in Table X. The earlier $N = 224$, $q = 160$ peak sweep is retained as the production-resolution audit in Table IX. The final column is the root-mean-square residual of the logarithmic high-frequency tail fit.

ℓ	retained	$M\omega_{\min}$	$\max \delta_0$	$\max \delta_1$	σ_{tail}
0	51	0.02	2.8×10^{-11}	6.9×10^{-12}	3.7×10^{-2}
1	40	0.12	1.9×10^{-11}	2.2×10^{-12}	6.7×10^{-2}
2	44	0.28	3.8×10^{-11}	1.5×10^{-12}	5.6×10^{-2}

TABLE IX. Production-resolution audit for the axisymmetric Kerr calculation. The entries compare the former production configuration $(N, q) = (224, 160)$ with the active reference configuration $(N, q) = (240, 192)$ at fixed $r_{\text{m}} = 40$. Each entry in the first four columns is $|X_{240,192} - X_{224,160}|/|X_{240,192}|$ for the indicated Green-function amplitude or flux coefficient; the final column is the absolute first-order balance residual of the reference run.

case	ΔA_{ref}	ΔA_H	$\Delta T^{(1)}$	$\Delta R^{(1)}$	$\lvert\delta_1\rvert_{\text{ref}}$
$\ell = 0,\ M\omega = .125$	4.5×10^{-7}	7.8×10^{-7}	2.3×10^{-6}	2.3×10^{-6}	2.7×10^{-12}
$\ell = 1,\ M\omega = .34$	5.9×10^{-6}	3.3×10^{-6}	1.5×10^{-5}	1.5×10^{-5}	2.2×10^{-12}
$\ell = 2,\ M\omega = .52$	9.6×10^{-6}	7.6×10^{-6}	3.1×10^{-5}	3.1×10^{-5}	1.2×10^{-12}

Each displayed point is computed independently at $(N, q) = (288, 320)$ and $(320, 384)$. The collocation equations are unchanged, but their rows and columns are equilibrated before the dense linear solve; this lowers the outer branch matrix condition number without changing the endpoint equations. Frequency-dependent matching radii lie between $12M$ and $30M$.

The primary acceptance condition for this survey is a relative change $\delta_N \chi_H \le 10^{-5}$ between the two resolutions, supplemented by a 10^{-8} linear-flux gate and a 10^{-10} Wronskian gate. Direct $R^{(1)}$ is retained as a separate diagnostic: 67 of the 77 displayed response points also satisfy $|T^{(1)} + R^{(1)}| \le 10^{-8}$ at both resolutions. The remaining ten points are not used to quote $R^{(1)}$; their horizon-side $T^{(1)}$ and χ_H pass the independent spectral gate, while the much smaller reflected interference correction is cancellation limited. No rejected response point is shown in Fig. 5. To avoid disconnected low-frequency islands, the $a/M = 0.5$ curves start at $x = 0.9$, corresponding to $M\omega = 0.121$ and 0.241 for $\ell = 1, 2$; the $a/M = 0.7$ and 0.9 curves start at $x = 0.6$, where their absolute frequencies are no lower than 0.123 and 0.188 for $\ell = 1$ and 0.245 and 0.376 for $\ell = 2$.

Figure 5(a) shows that $T^{(1)} < 0$ below $x = 1$ and $T^{(1)} > 0$ above it for all six families. Panel (b) shows why this sign change is regular rather than discontinuous: χ_H is positive, finite, and smooth through the threshold. We evaluate the threshold limit by symmetrizing the pairs $x = 1 \pm 0.001$ and 1 ± 0.01 and Richardson-extrapolating the leading even error. Table XIV lists the resulting $\chi_H^\star$ and the independently finite slope $\lim_{p\to 0} T^{(1)}/p$. The two spectral resolutions agree in $\chi_H^\star$ to between 4.2×10^{-9} and 2.8×10^{-7}.

The positive sign has an immediate flux interpretation. In the convention of Eq. (5), $\epsilon > 0$ gives $T^{(1)} < 0$ and $R^{(1)} = -T^{(1)} > 0$ for $p < 0$, so the weak interaction increases the superradiant amplification by the fractional amount $\epsilon|\mathcal{A}_{\text{in}}|^2\chi_H$. A conventional positive quartic potential has $\epsilon = -\lambda$ in our action convention and therefore suppresses the amplification at this order. This is a perturbative statement; it does not describe nonlinear saturation.

The response also develops a mode-dependent maximum above threshold at high spin. For $a/M = 0.9$, the largest sampled values are $\chi_H = 0.369248$ at $(\ell, m, x) = (1, 1, 1.30)$ and $\chi_H = 0.378482$ at $(2, 2, 1.20)$, with resolution changes 7.7×10^{-7} and 1.5×10^{-6}, respectively. Moving the matching radius from its reference value $12M$ to $10M$ and $16M$ changes these peak values by only 9.4×10^{-8} and 2.4×10^{-7}. The response then falls rapidly; according to the same 10^{-5} relative gate, the plotted high-spin curves are truncated at $x = 1.60$ for $\ell = 1$ and $x = 1.30$ for $\ell = 2$. We regard these as resolved real-frequency response maxima, not as quasinormal-mode pole fits.

C. Near-extremal scaling and multipole reordering

The growth visible between $a/M = 0.5$ and 0.9 does not continue as an arbitrary function of spin. To identify its geometric scale, we introduce the surface gravity and near-threshold variables

$$\kappa = \frac{r_+ - r_-}{2(r_+^2 + a^2)}, \qquad y = \frac{p}{\kappa}, \qquad \Xi = M\kappa\chi_H. \tag{91}$$

TABLE X. High-frequency convergence audit for the active tail data. The quadrature columns compare $q = 1792$ with the $q = 2048$ reference at $N = 240$; the radial-order column compares $N = 224$ with $N = 240$ at $q = 2048$. All entries except the final balance residual are relative changes with respect to the reference.

case	$\Delta T^{(1)}[q]$	$\Delta T^{(1)}[N]$	$\Delta A_{\rm ref}[q]$	$\Delta A_H[q]$	$\lvert\delta_1\rvert_{\rm ref}$
$\ell = 0,\ M\omega = .52$	3.0×10^{-5}	7.1×10^{-10}	2.3×10^{-5}	1.6×10^{-9}	7.8×10^{-15}
$\ell = 1,\ M\omega = .77$	1.7×10^{-4}	1.0×10^{-9}	3.1×10^{-4}	1.2×10^{-9}	2.7×10^{-13}
$\ell = 2,\ M\omega = .92$	2.5×10^{-4}	9.5×10^{-8}	3.4×10^{-4}	2.5×10^{-9}	1.0×10^{-14}

TABLE XI. Method-level convergence stress tests for the scalar Kerr Green integrals. Each entry is the maximum relative change in the indicated amplitude over the non-reference runs. The N scan compares $N = 224, 288$ with $N = 256$; the matching scan compares the listed alternatives with the reference $r_{\rm m}$; and the quadrature scan compares lower orders with $q = 192$. These nonaxisymmetric self-channel tests are validation data for the numerical method and are not included in the axisymmetric production curves.

case	$\Delta A_{\rm ref}[N]$	$\Delta A_H[N]$	$\Delta A_{\rm ref}[r_{\rm m}]$	$\Delta A_H[r_{\rm m}]$	$\Delta A_{\rm ref}[q]$	$\Delta A_H[q]$	$\max \delta_W$
$a = .5,\ M\omega = .24$	2.1×10^{-5}	1.1×10^{-7}	3.3×10^{-5}	5.4×10^{-8}	2.8×10^{-8}	2.5×10^{-8}	5.9×10^{-16}
$a = .5,\ M\omega = .30$	3.8×10^{-5}	2.8×10^{-8}	2.0×10^{-5}	4.4×10^{-8}	1.8×10^{-6}	1.7×10^{-6}	3.7×10^{-16}
$a = .9,\ M\omega = .58$	1.5×10^{-6}	1.2×10^{-8}	1.7×10^{-6}	9.4×10^{-9}	1.4×10^{-10}	5.1×10^{-7}	2.9×10^{-16}

TABLE XII. Angular-discretization audit. Each entry is the largest relative change over $(\ell, m, M\omega; \ell') = (2, 0, 1.0; 2)$ and $(2, 2, 0.30; 6)$, measured against the reference $(L_{\rm extra}, q_\theta) = (24, 512)$.

setting $(L_{\rm extra}, q_\theta)$	$\max \Delta\lambda$	$\max \Delta C^{(0)}$	$\max \Delta C^{(2)}$
active $(16, 256)$	4.4×10^{-16}	3.7×10^{-14}	5.2×10^{-14}
basis $(12, 512)$	4.4×10^{-16}	1.6×10^{-16}	0
quadrature $(24, 128)$	0	1.5×10^{-14}	2.0×10^{-14}

The two limiting procedures must be distinguished before fitting. Our threshold data mean the iterated limit

$$\chi_H^\star(\kappa) = \lim_{y\to 0} \chi_H(\kappa, y), \qquad \text{followed by} \qquad \kappa \to 0, \tag{92}$$

whereas a fixed-y sequence takes $\kappa \to 0$ and $p = y\kappa$ together. For each spin and mode we sample the four threshold offsets $y = (-0.05, -0.025, 0.025, 0.05)$ and form

$$\begin{aligned} \overline{\chi}(\eta) &= \frac{\chi_H(+\eta) + \chi_H(-\eta)}{2}, \\ \chi_H^\star &= \frac{4\overline{\chi}(0.025) - \overline{\chi}(0.05)}{3}. \end{aligned} \tag{93}$$

The symmetrization removes the odd terms and the second combination removes the leading $O(y^2)$ error. Thus the label $p = 0$ below denotes an ingoing limit inferred from four nonzero-p solves, not a solve at the degenerate Frobenius point. The radial domains still contain $r = r_+$ and impose the regular endpoint equation there; no horizon cutoff is introduced.

The common grid contains $a/M = (0.98, 0.99, 0.995, 0.997, 0.999)$ and modes $(\ell, m) = (1,1), (2,1), (2,2), (3,2), (3,3), (4,4), (5,5)$. We add $a/M = 0.9995, 0.9999$ for $(1,1), (2,1), (2,2), (3,2)$. In addition to the four threshold offsets, independent fixed-y rows are evaluated at $y = -0.5, 0.5, 1$. The base resolutions are $(N, q) = (144, 144)$ and $(176, 176)$; the most cancellation-sensitive supplementary channels are repeated at $(224, 224)$ and $(256, 256)$. A row is admitted only when the paired response changes by at most 5×10^{-4}, the linear-flux residual is at most 2×10^{-8}, and the Wronskian error is at most 10^{-12}. Forty threshold limits and 126 fixed-y points pass. Among accepted rows the largest corresponding values are 1.21×10^{-4}, 1.60×10^{-11}, and 7.05×10^{-16} at threshold, and 4.92×10^{-4}, 6.94×10^{-9}, and 5.59×10^{-16} at fixed y. Failed extreme supplementary rows are omitted from all fits and figures.

We fit each path independently to

$$\chi_H = A(M\kappa)^{-\alpha} \tag{94}$$

by ordinary least squares in $(\ln M\kappa, \ln \chi_H)$. The fit is repeated over all accepted spins, the nearest four, and the nearest three. We call a mode inverse-κ only if the nearest-three exponent lies in $0.8 \le \alpha \le 1.2$, has $R^2 \ge 0.995$, and changes by at most 0.1 from the nearest-four fit. Small drifting exponents are additionally compared with the finite-limit model $C_0 + C_1(M\kappa)^\beta$.

There is a useful linear-theory classification. In the extremal angular convention of Ref. [47],

$$\begin{aligned} h_{\ell m} &= \frac{1}{2} + \sqrt{\frac{1}{4} + K_{\ell m} - 2m^2}, \\ K_{\ell m} &= A_{\ell m}(m/2) + \frac{m^2}{4}. \end{aligned} \tag{95}$$

Negative square-root discriminant defines the principal, complex-h family; positive discriminant defines the sup-

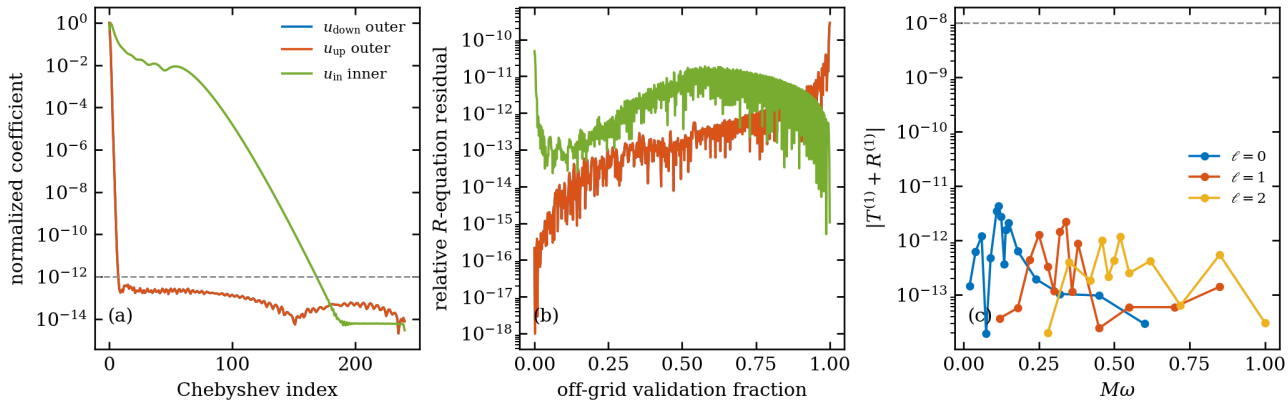


FIG. 4. Internal numerical accuracy diagnostics for the axisymmetric Kerr calculation. Panels (a) and (b) use the representative peak point $a = 0.5$, $m = 0$, $\ell = 2$, and $M\omega = 0.52$ with $N = 240$ and quadrature order $q = 192$. Panel (a) shows the normalized Chebyshev coefficients of the phase-peeled homogeneous solutions on the outer and inner subdomains. Panel (b) shows the relative residual (55) after evaluating the spectral interpolant and its analytic derivatives on 600 off-grid Chebyshev–Gauss points per branch in the original radial Teukolsky equation. Panel (c) shows the first-order flux-balance residual for the retained axisymmetric points. The dashed line marks the acceptance threshold 10^{-8} used before any point enters the manuscript figures or fits.

TABLE XIII. Representative Kerr target-channel matrix for an incident $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$ mode. The angular entries are absolute values of the two source projectors in Eq. (29); the Green amplitudes are unit-horizon first-order amplitudes. The odd channels vanish within the displayed precision.

ℓ'	$\lvert C^{(0)}_{\ell'\ell m}\rvert$	$\lvert C^{(2)}_{\ell'\ell m}\rvert$	$\lvert A^{(1)}_{\mathrm{ref},\ell'\lvert\ell m}\rvert$	$\lvert A^{(1)}_{H,\ell'\lvert\ell m}\rvert$
2	1.136×10^{-1}	1.034×10^{-2}	1.137×10^{6}	3.804×10^{3}
3	0	0	0	0
4	3.578×10^{-2}	5.513×10^{-3}	1.362×10^{4}	1.319
5	0	0	0	0
6	5.341×10^{-3}	5.364×10^{-3}	2.265×10^{3}	5.383×10^{-7}

plementary, real-h family. This is a classification of the associated linear extremal Green function, not by itself a derivation of the nonlinear exponent.

Table XV gives the nearest-three exponents. Every tested principal mode, $(2, 2)$ through $(5, 5)$, is in the inverse-κ class on all four paths. Their threshold exponents span 1.038–1.064; at fixed $y = -0.5, 0.5, 1$ the respective ranges are 1.027–1.051, 1.044–1.091, and 1.042–1.121. The smallest R^2 among these 16 fits is 0.99987. Extending $(2, 2)$ to seven spins gives

$$\chi^{\star}_{H,22} = 0.02872(M\kappa)^{-1.0037}, \qquad R^2 = 0.99975, \quad (96)$$

The nominal ordinary-least-squares 95% interval for the exponent is $[0.9855, 1.0219]$, the largest leave-one-out relative prediction error is 5.02%, and the largest paired-resolution response change is 1.26×10^{-5}. The interval is descriptive: it quantifies residuals within the seven resolved points but does not include model uncertainty from the two-branch matching function. This fit motivates the resolved leading form

$$\chi^{\star}_{H,22} \simeq \frac{0.029}{M\kappa} = \frac{0.0046}{MT_H}, \qquad T_H = \frac{\kappa}{2\pi}. \quad (97)$$

The numerical split is sharper than the statement that corotating modes are enhanced. The fully corotating dipole $(1, 1)$ has real h, and its threshold exponent moves from 0.498 over all seven spins to 0.445 over the nearest four and 0.415 over the nearest three. We therefore do not identify an asymptotic fractional power. For $(2, 1)$ and $(3, 2)$ the threshold exponents similarly decrease from 0.124 to 0.092 and from 0.312 to 0.247 as the window is moved inward. The finite-limit model is favored over a pure power by $\Delta\mathrm{AICc} = 18.3$ and 19.8, respectively, where positive values favor the finite model. These are bounded candidates, while $(1, 1)$ remains a resolved crossover. Empirically, the robust $1/\kappa$ response follows the principal conformal sector rather than the condition $m = \ell$ alone.

The enhancement is a boundary-layer phenomenon, not only a threshold number. For $a/M = 0.98, 0.99, 0.995$ we scan $-0.5 \le y \le 1.2$ for $\ell = m = 2, 3$. All 60 pairs pass the $N = 144$–176 comparison; the largest relative response change is 1.7×10^{-4}, the largest linear and first-order flux-balance residuals are 1.1×10^{-8} and 8.1×10^{-9}, and the largest Wronskian error is 4.9×10^{-16}. As shown

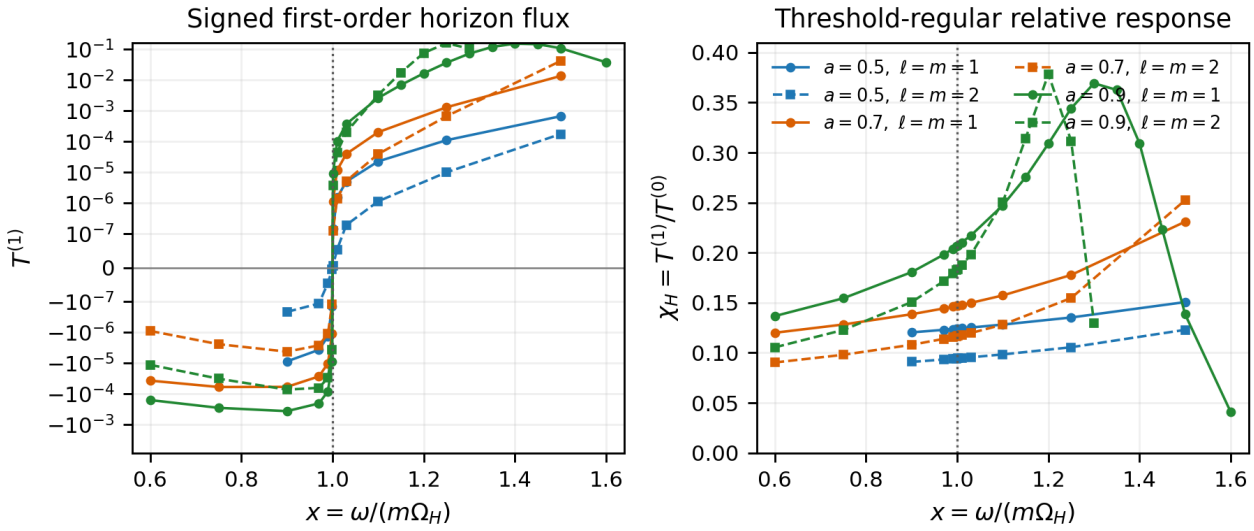


FIG. 5. Accepted nonaxisymmetric scalar Kerr response for $a/M = 0.5, 0.7, 0.9$ and $\ell = m = 1, 2$. The dotted vertical line is the superradiant threshold $x = 1$. (a) The signed first-order horizon coefficient vanishes and changes sign at the threshold. (b) The ratio $\chi_H = T^{(1)}/T^{(0)}$ remains finite and positive, increases with spin near threshold, and develops mode-dependent maxima for $a/M = 0.9$. Only points passing the two-resolution response gate are displayed; each curve is truncated before its cancellation-limited low- or high-response tail.

TABLE XIV. Threshold-regular nonlinear horizon response. The limit is obtained from the symmetric pairs $x = 1 \pm 0.001$ and 1 ± 0.01 with leading-even-error Richardson extrapolation. The last column is the relative change between the $(N, q) = (288, 320)$ and $(320, 384)$ extrapolations.

a/M	$\ell = m$	$\chi_H^\star$	$\lim_{p\to 0} T^{(1)}/p$	$\delta_N \chi_H^\star$
0.5	1	0.124051	1.0573×10^{-3}	4.2×10^{-9}
0.5	2	0.094350	1.8295×10^{-5}	2.8×10^{-7}
0.7	1	0.146970	5.3226×10^{-3}	1.5×10^{-8}
0.7	2	0.116670	2.9971×10^{-4}	8.7×10^{-9}
0.9	1	0.206963	2.9206×10^{-2}	7.9×10^{-8}
0.9	2	0.183251	5.8233×10^{-3}	6.4×10^{-8}

in Fig. 9, the data support the numerical scaling form

$$\chi_{H,\ell\ell}(p, a) = \frac{1}{M\kappa} F_\ell^{\rm num}\left(\frac{p}{\kappa}\right)[1 + \delta_\ell(\kappa, p/\kappa)], \qquad \ell = 2, 3, \tag{98}$$

For $-0.5 \le y \le 1$, the relative scatter across the three spins is at most 3.9% for $\ell = 2$ and 3.7% for $\ell = 3$. A denser grid with $\Delta y = 0.05$ locates the sampled maxima at $y = 0.80$–0.90 and gives $\Xi_{\rm pk} = 0.06419$ with 2.22% relative scatter among the four $(a/M, \ell) = (0.99, 2), (0.99, 3), (0.995, 2), (0.995, 3)$ cases. Thus the resolved peak-height envelope grows as $1/\kappa$ while its frequency width scales as κ; Eq. (98) is an empirical collapse over the stated spin window, not an asserted exact-extremal expansion.

The fixed-y columns of Table XV show that the modal split persists away from the single threshold number. In particular, the principal modes retain near-unit exponents on both sides of the bound, while the supplementary controls have exponents that shrink as the fit window moves inward. This comparison ties the enhancement to the principal near-superradiant throat sector rather than to generic growth of the cubic source with spin.

1. *Kerr/CFT interpretation: from linear absorption to nonlinear order locking*

The collapse above can be organized by the finite-temperature throat description without using it as an input to the numerical calculation. The standard left- and right-moving Kerr/CFT temperatures are

$$T_L = \frac{r_+ + r_-}{4\pi a}, \qquad T_R = \frac{r_+ - r_-}{4\pi a} = \frac{\kappa(r_+^2 + a^2)}{2\pi a}. \tag{99}$$

Thus $T_L \to 1/(2\pi)$ while $T_R = \mathcal{O}(M\kappa)$ in the extremal limit. The near-superradiant right-moving fre-

TABLE XV. Modal near-extremal scaling audit. The $y = 0$ column is the iterated threshold limit in Eqs. (92) and (93); the other columns hold the displayed nonzero y fixed. Each α_y is a nearest-three effective logarithmic slope, so its precise spin endpoints are channel dependent and are recorded in the accompanying CSV file. These numbers diagnose window drift; no asymptotic power is assigned to a supplementary mode.

(ℓ, m)	$h_{\ell m}$	$\alpha_{-0.5}$	α_0	$\alpha_{0.5}$	α_1	numerical class
$(1,1)$	1.171	0.471	0.415	0.308	0.236	drifting crossover
$(2,1)$	2.596	0.103	0.092	0.041	0.000	finite-limit favored
$(2,2)$	$0.5 + 0.946i$	1.027	1.038	1.044	1.121	κ^{-1} envelope
$(3,2)$	2.717	0.218	0.247	0.125	0.090	finite-limit favored
$(3,3)$	$0.5 + 1.938i$	1.036	1.051	1.071	1.042	κ^{-1} envelope
$(4,4)$	$0.5 + 2.850i$	1.047	1.060	1.085	1.052	κ^{-1} envelope
$(5,5)$	$0.5 + 3.741i$	1.051	1.064	1.091	1.101	κ^{-1} envelope

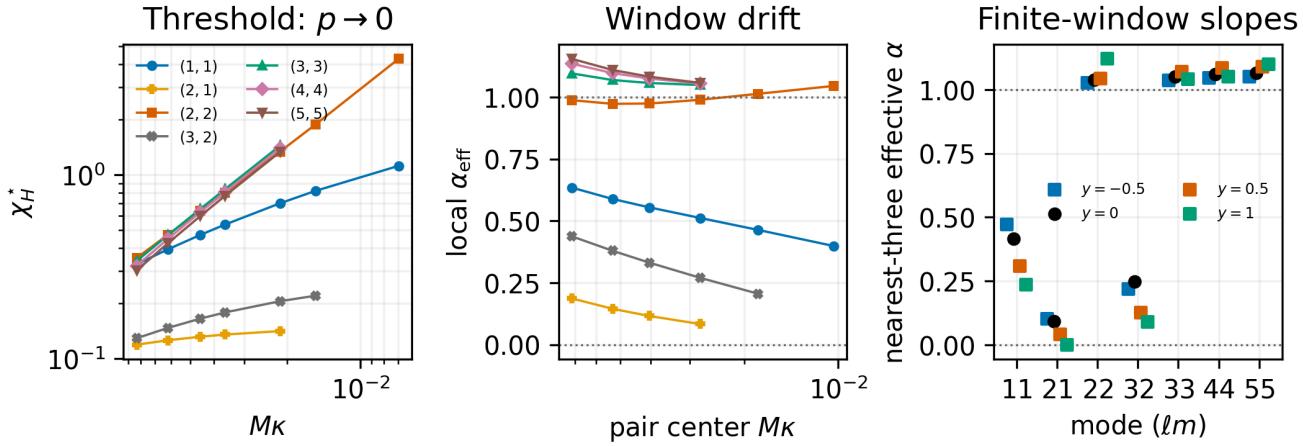


FIG. 6. Path-controlled modal scaling audit. (a) Threshold responses obtained from the four-point $y \to 0$ extrapolation. (b) Adjacent-pair effective exponents; the principal $(2,2)$–$(5,5)$ curves approach the $\alpha = 1$ guide, while the supplementary curves drift downward. (c) Nearest-three effective slopes at the iterated threshold ($y = 0$) and at three fixed nonzero detunings. The supplementary points are finite-window diagnostics, not h-derived critical exponents. Only samples passing the paired-resolution, linear-flux, and Wronskian gates are shown.

quency obeys $\varpi_R \equiv \omega_R/(2\pi T_R) \propto p/\kappa = y$; convention-dependent constant factors can be absorbed into the definition of ω_R. The angular eigenvalue enters through the conformal weight in Eq. (95). These are precisely the variables in which the linear near-region absorption probability is reproduced by a thermal two-dimensional correlator [42–44].

The CFT two-point function determines the near-region factor, but the quantity normalized at infinity also contains both near–far matching branches. In the matched scalar solution their coefficients scale as $A \sim (M\kappa)^{1-h}$ and $B \sim (M\kappa)^h$, while the incident amplitude is proportional to $A\mathcal{D}_{+,h} + B\mathcal{D}_{-,h}$ and the horizon flux carries one power of κ. Factoring the A branch gives, up to κ-independent normalizations, the leading form [42]

$$T^{(0)}_{\ell m}(\kappa, y) = \frac{(M\kappa)^{2\,\mathrm{Re}\,h-1}\mathcal{N}_h(y)}{\left|\mathcal{D}_{+,h}(y) + \mathcal{D}_{-,h}(y)(M\kappa)^{2h-1}\right|^2}\,[1 + o(1)]. \tag{100}$$

Here $\mathcal{D}_{\pm,h}$ include the far-zone matching coefficients, and $\mathcal{N}_h$ contains the signed horizon factor. Equation (100), rather than the isolated CFT power, is the appropriate linear input for the present asymptotically flat problem. If h is real, the second matching branch is suppressed and $T^{(0)} \sim (M\kappa)^{2h-1}$. If $h = 1/2 + i\nu$, the two branches have the same magnitude and $(M\kappa)^{2h-1} = \exp[2i\nu \ln(M\kappa)]$. The linear absorption then carries no net real power of κ, but is generally a bounded log-periodic matching function away from isolated resonant zeros. It need not approach a κ-independent constant.

The crucial step from a linear correlator to the nonlinear susceptibility is the fixed-incident normalization. The unit-horizon solution becomes a unit-incident solution after multiplication by $1/B^{\rm inc}$. A useful dimensionless measure of the resulting horizon intensity is

$$\mathcal{I}_H \equiv \frac{M^2}{|B^{\rm inc}|^2} = \frac{M^2\omega T^{(0)}}{p(r_+^2 + a^2)}. \tag{101}$$

At fixed nonzero y, $p = y\kappa$, so Eq. (100) implies the linear

intensity scaling

$$\mathcal{I}_H \sim \begin{cases} (M\kappa)^{2h-2}\mathcal{I}_h(y), & h \in \mathbb{R}, \\ (M\kappa)^{-1}\mathcal{Q}_h(y, \ln M\kappa), & h = 1/2 + \mathrm{i}\nu, \end{cases} \tag{102}$$

where $\mathcal{Q}_h$ retains the two-branch log-periodic matching. The power $2h-2$ in the first line belongs to the *linear fixed-incident horizon intensity*; it is not an exponent for the nonlinear susceptibility. For a principal mode this does not say that the horizon energy flux diverges. Its kinematic factor p vanishes; the field intensity required to carry a fixed incident flux through that increasingly soft channel grows. At the threshold itself, Eq. (102) means first taking the fixed-y scaling limit and then continuing the matching function to $y = 0$, exactly in the order used by Eq. (92). No division by a numerical $0/0$ is performed.

Equation (74) now gives the factorization needed to compare operator orders. Strip the incident normalization from the response by defining the dimensionless unit-horizon nonlinear kernel

$$\mathcal{K}_{n,\ell m}(\kappa, y) \equiv \frac{M^{2n-2}\chi^{(n)}_{H,\ell m}}{\mathcal{I}^n_H} = \frac{2\,\mathrm{Re}\,A^{(1)}_{H,n}}{M^2}. \tag{103}$$

The equality follows directly from Eq. (74); it is exact in the conventions of this paper. The kernel contains the projected unit-horizon Green integral and the nonlinear near–far overlap, but no power of the incident normalization. Combining Eqs. (102) and (103) gives the logically complete factorization

$$\begin{aligned} M^{2n-2}\chi^{(n)}_H &\sim (M\kappa)^{n(2h-2)}\mathcal{I}^n_h(y)\mathcal{K}_{n,h}(\kappa, y), && h \in \mathbb{R}, \\ M^{2n-2}\chi^{(n)}_H &\sim (M\kappa)^{-n}\mathcal{Q}^n_h(y, \ln M\kappa)\mathcal{K}_{n,h}(\kappa, y), && h = 1/2 + \mathrm{i}\nu. \end{aligned} \tag{104}$$

Only the explicit linear-normalization factor in this equation is fixed by the known matched linear theory. The nonlinear kernel is not fixed by the conformal weight. Consequently the principal-sector envelope $\chi^{(n)}_H = \mathcal{O}(\kappa^{-n})$ follows conditionally if $\mathcal{Q}^n_h\mathcal{K}_{n,h}$ remains bounded and carries no additional real power. It may still contain logarithmic or log-periodic modulation. This is a power-counting argument, not an analytic evaluation of the nonlinear NHEK integral.

We audit this factorization directly at $y = 0.5$ using the recorded B^{inc}. For any quantity X in Table XVI, α_X denotes a finite-window effective slope $X \propto \kappa^{-\alpha_X}$. The exact definition gives $\alpha_\chi = n\alpha_\mathcal{I} + \alpha_\mathcal{K}$ within rounding. The principal $(3,3)$ kernel has a nearly vanishing effective slope, whereas the $(2,2)$ kernel retains a 0.10–0.13 drift that can include unresolved two-branch modulation. Thus the full response collapse is stronger than an independent numerical demonstration that $\mathcal{K}_{n,h}$ has a strict constant limit. In the supplementary rows the kernel slope is large and compensates the vanishing linear intensity, explicitly ruling out an h-only formula for the response.

In a boundary-source language, the distinction can be written schematically as

$$\langle\mathcal{O}\rangle_J = G^{(2)}_R J + \epsilon_n \int G^{(2n+2)}_{R,\mathrm{conn}} J^{n+1}(J^*)^n + \cdots . \tag{105}$$

The first term controls the linear throat normalization. The second is the contact contribution associated with the bulk $|\Phi|^{2n+2}$ vertex; for the cubic equation, $n = 1$, it is a four-point rather than a three-point response. Kerr/CFT three-point calculations demonstrate how bulk interactions encode higher-point data [45], but the real-time correlator and asymptotically flat matching needed for Eq. (105) have not been computed here. Consequently linear Kerr/CFT fixes neither the sign nor the normalization of $\mathcal{F}_{n,h}(y)$. Those quantities, including the positive threshold response and the peak near $y \simeq 0.8$–0.9, are genuine outputs of the nonlinear Green-function experiment.

For supplementary modes the same equation explains why no h-only response law exists. Linear theory supplies the factor $(M\kappa)^{n(2h-2)}$, but $\mathcal{K}_{n,h}$ can carry a competing power because the nonlinear source samples the throat entrance and far matching region. Therefore neither $2\,\mathrm{Re}\,h - 2$ nor $n(2\,\mathrm{Re}\,h - 2)$ is an exponent of χ_H unless the kernel is separately shown to be asymptotically constant. Our data do not show that. The drifting $(1,1)$ slopes and the $(2,1), (3,2)$ crossovers are reported only as finite-window behavior, not as supplementary-sector critical exponents.

2. *Higher local powers and operator-order hierarchy*

The monopole hierarchy in Eq. (85) does not determine what a higher local power does to a nonaxisymmetric throat mode. We therefore repeat the path-controlled experiment for $n = 2, 3$, using the finite-window fit model

$$\begin{aligned} \widehat{\chi}^{(n)}_{H,\ell m} &\equiv M^{2n-2}\chi^{(n)}_{H,\ell m}, \\ \widehat{\chi}^{(n)}_{H,\ell m} &= A_{n,\ell m}(y)(M\kappa)^{-\alpha_{n,\ell m}(y)}. \end{aligned} \tag{106}$$

The modes $(1,1), (2,1), (2,2), (3,3)$ provide low- and high-real-weight supplementary controls and two principal channels. The common spin grid is $a/M = (0.98, 0.99, 0.995, 0.997, 0.999)$; every mode is additionally computed at $a/M = (0.9995, 0.9998, 0.9999)$. We use the same four nonzero threshold offsets as Eq. (93) and an independent fixed-$y = 0.5$ sequence. For $(1,1)$ and $(2,2)$ the wider fixed-y check also includes $y = -0.5$ and 1. The base resolution pair is $(N, q) = (144, 144)$ and $(176, 176)$, while the first three added spins use $(224, 224)$ and $(256, 256)$. For both principal modes we further add $a/M = (0.99995, 0.999975, 0.9999875, 0.99999375)$ at $N = q = 256, 288$ and $a/M = (0.999996875, 0.9999984375)$ at $N = q = 288, 320$. The resulting primary fit window contains

TABLE XVI. Fixed-$y = 0.5$ factorization audit. The principal and $(1,1)$ rows use the nearest seven accepted spins; the shorter $(2,1)$ sequence uses its five accepted spins. All fitted columns are effective slopes in the convention $X \propto \kappa^{-\alpha_X}$. The linear column is the matched-asymptotic intensity exponent: $2-2h$ for real h, and the principal-sector envelope value 1 for complex h. The last four columns come from the accepted numerical rows and satisfy $\alpha_\chi = n\alpha_\mathcal{I} + \alpha_\mathcal{K}$. They diagnose the factors in Eq. (104); they are not supplementary-sector asymptotic exponents.

(ℓ, m)	$\alpha_\mathcal{I}^{\rm lin}$	$\alpha_\mathcal{I}^{\rm fit}$	$\alpha_{\mathcal{K},2}$	$\alpha_{\chi,2}$	$\alpha_{\mathcal{K},3}$	$\alpha_{\chi,3}$
$(1,1)$	-0.341	-0.023	0.724	0.677	1.059	0.988
$(2,1)$	-3.192	-2.783	5.829	0.263	8.800	0.451
$(2,2)$	1 (envelope)	0.973	0.099	2.045	0.129	3.048
$(3,3)$	1 (envelope)	1.013	-0.003	2.023	-0.003	3.036

seven points with $0.9999 \le a/M \le 0.9999984375$. Acceptance requires paired-response convergence, linear balance, Wronskian constancy, and either absolute or flux-relative first-order balance.

The principal modes exhibit a numerical order-locked envelope:

$$\widehat{\chi}^{(n)}_{H,\ell\ell}(\kappa, y) = \frac{\mathcal{F}^{\rm num}_{n,\ell}(y, \ln M\kappa)}{(M\kappa)^n}, \qquad \mathcal{F}^{\rm num}_{n,\ell} = \mathcal{O}(1) \quad \text{on the resolved paths.} \tag{107}$$

As shown in Tables XVII and XVIII and Fig. 7, the seven-point exponents at the iterated threshold are 2.0587 and 3.0567 for the $(2,2)$ mode at $n = 2, 3$, and 2.0257 and 3.0408 for $(3,3)$. At fixed $y = 0.5$ they are 2.0447, 3.0475, 2.0226, and 3.0357, in the same order. The minimum principal R^2 is 0.9999766, and the largest leave-one-out relative prediction error is 2.30%. Nominal ordinary-least-squares intervals resolve modest offsets from $\alpha_n = n$ in most channels. These intervals summarize finite-window residuals and are not stochastic confidence bounds on an asymptotic model. The data therefore support an order-locked leading envelope with bounded residual modulation, not an exact pure power. Thus, over the accepted spin windows, adding one factor $|\Phi|^2$ to the local source adds one resolved inverse power of the surface gravity to the fixed-incident response. Equation (104) supplies conditional analytic power counting for this envelope, while Table XVI shows that the linear intensity and nonlinear kernel need not be separately constant.

The collapse test is more restrictive than fitting slopes independently. Over the seven-point primary window, the threshold means of $(M\kappa)^n \widehat{\chi}^{(n)}_H$ are 4.0128×10^{-4} and 5.6432×10^{-6} for $(2,2)$ at $n = 2, 3$, with relative standard deviations 4.06% and 3.95%. For $(3,3)$ the corresponding means are 1.0660×10^{-3} and 4.0687×10^{-5}, with relative standard deviations 1.83% and 2.88%. The nonzero scatter is the same finite-window modulation resolved by the free-power fits. These order-dependent constants decrease rapidly, but the extra inverse powers of κ dominate sufficiently close to extremality.

The nested-window audit gives the same hierarchy without using the seven-point interval as the only diagnostic. Moving from the nearest seven to the nearest three spins changes the principal $n = 2$ slopes from 2.0587 to 2.0405 for $(2,2)$ and from 2.0257 to 2.0121 for $(3,3)$ at threshold. For $n = 3$ the corresponding nearest-three values are 3.0485 and 3.0192. At fixed $y = 0.5$, all four nearest-three slopes lie between 2.0104 and 2.0160 or between 3.0164 and 3.0196, according to operator order. This seven-to-three-point comparison supports the order hierarchy, but the three-point fits have only one residual degree of freedom and are used only as a window-drift diagnostic.

The supplementary sector does not obey a common power. At threshold the low-real-weight $(1,1)$ effective slope increases from the cubic value 0.415 to 0.937 and 1.402 for $n = 2, 3$, but remains well below the order-locked values. It is also strongly path dependent: at $y = 0.5$ the latter slopes fall to 0.677 and 0.988. The $(2,1)$ control grows much more weakly in the accepted five-spin window. Its three added-spin pairs fail only the response-resolution gate, with no accompanying loss of linear balance or Wronskian constancy, and are excluded from Table XVII and all fits. We therefore identify a resolved crossover, not an asymptotic power or a proof of a finite limit. The data establish an order-locked envelope in the principal sector and its absence in two distinct supplementary controls. No supplementary exponent is predicted from h alone.

Across 480 paired higher-power samples, 82 threshold limits and 122 fixed-y values pass all gates. Among accepted data, the largest response change is 4.6×10^{-6} at threshold and 3.4×10^{-5} at fixed y; the largest linear-balance residuals are 1.6×10^{-11} and 7.0×10^{-9}, and the largest Wronskian error is 5.8×10^{-16}. Within the seven-point principal window the largest paired response changes are 4.5×10^{-7} at threshold and 8.9×10^{-6} at fixed y, while the largest flux-relative first-order balance residual is 6.7×10^{-9}. Recomputing all $n = 2, 3$ angular projectors at the hardest spin with 24 extra spherical modes and 512 Gauss points changes them by at most 4.1×10^{-15}.

TABLE XVII. Higher-power modal scaling. The displayed α_n are finite-window effective slopes of Eq. (106). The $p \to 0$ column uses the four-point ingoing extrapolation; the other holds $y = p/\kappa = 0.5$. The principal rows use the seven-point primary window, $(1,1)$ uses its seven nearest accepted spins, and $(2,1)$ uses its five accepted spins. The supplementary entries are not asserted asymptotic powers.

(ℓ, m)	$h_{\ell m}$	n	$\alpha_n(p \to 0)$	$\alpha_n(y = 0.5)$	numerical class
$(1,1)$	1.171	2	0.937	0.677	drifting slope
		3	1.402	0.988	drifting slope
$(2,1)$	2.596	2	0.468	0.263	weak crossover
		3	0.812	0.451	weak crossover
$(2,2)$	$0.5 + 0.946i$	2	2.059	2.045	locked envelope
		3	3.057	3.048	locked envelope
$(3,3)$	$0.5 + 1.938i$	2	2.026	2.023	locked envelope
		3	3.041	3.036	locked envelope

TABLE XVIII. Seven-point principal-mode fit audit over $0.9999 \le a/M \le 0.9999984375$. The interval is the nominal 95% ordinary-least-squares interval for the free logarithmic slope. It is a finite-window residual diagnostic, not a confidence bound on the asymptotic model. The leave-one-out (LOO) column is the largest relative prediction error, and δ_N is the largest paired-resolution response change. These statistical intervals do not include model uncertainty from bounded two-branch modulation.

(ℓ, m)	n	path	α_n [95% interval]	R^2	LOO max	δ_N
$(2,2)$	2	$p \to 0$	2.0587 [2.0519,2.0655]	0.99999180	1.28%	4.50×10^{-7}
$(2,2)$	2	$y = 0.5$	2.0447 [2.0334,2.0561]	0.99997661	2.18%	8.85×10^{-6}
$(2,2)$	3	$p \to 0$	3.0567 [3.0482,3.0652]	0.99999420	1.46%	4.32×10^{-8}
$(2,2)$	3	$y = 0.5$	3.0475 [3.0373,3.0578]	0.99999141	2.02%	1.61×10^{-6}
$(3,3)$	2	$p \to 0$	2.0257 [2.0178,2.0337]	0.99998836	1.43%	4.20×10^{-7}
$(3,3)$	2	$y = 0.5$	2.0226 [2.0154,2.0298]	0.99999031	1.31%	7.56×10^{-6}
$(3,3)$	3	$p \to 0$	3.0408 [3.0281,3.0535]	0.99998681	2.30%	3.81×10^{-8}
$(3,3)$	3	$y = 0.5$	3.0357 [3.0242,3.0472]	0.99998915	2.09%	8.95×10^{-7}

3. *Finite-bandwidth hierarchy and peak migration*

The pointwise hierarchy has a direct consequence for a fixed interval in the throat frequency. Define the dimensionless windowed spectral weight

$$\mathcal{W}_{n,\ell m}[y_-, y_+] \equiv \int_{m\Omega_H + y_-\kappa}^{m\Omega_H + y_+\kappa} \widehat{\chi}^{(n)}_{H,\ell m}(\omega)\, d(M\omega). \quad (108)$$

Since $d(M\omega) = M\kappa\, dy$, Eq. (107) gives

$$\mathcal{W}_{n,\ell\ell}[y_-, y_+] = (M\kappa)^{1-n} \int_{y_-}^{y_+} \mathcal{F}^{\mathrm{num}}_{n,\ell}(y, \ln M\kappa)\, dy. \quad (109)$$

Thus an integration over a thermal-width interval removes one inverse power of surface gravity. It makes the cubic spectral weight finite at leading power, but leaves the $n > 1$ weights enhanced as κ^{1-n} if the integrated scaling function remains bounded and nonzero. This is a windowed integral of monochromatic susceptibilities; it is not a nonlinear wave-packet calculation, which would also contain frequency mixing and a specified incident spectral envelope.

We test Eq. (109) on the common ten-point grid

$$y = (-0.5, -0.2, -0.05, 0.05, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2) \quad (110)$$

for both principal modes. The cubic calculation uses $a/M = (0.98, 0.99, 0.995)$ and $(N, q) = (144, 144), (176, 176)$; the $n = 2, 3$ calculation uses $a/M = (0.995, 0.9999, 0.99999375)$ and $(N, q) = (256, 256), (288, 288)$. Trapezoidal integration is applied identically to each paired grid. All 180 monochromatic rows pass the response, flux, and Wronskian gates. The largest pointwise resolution change is 4.09×10^{-4}, the largest integral change is 7.78×10^{-5}, and the largest flux-relative first-order balance residual is 1.54×10^{-8}.

Writing $\mathcal{W}_n \propto (M\kappa)^{-\beta_n}$ over these three spins, the effective exponents are

	$\beta_{n,22}$	R^2_{22}	$\beta_{n,33}$	R^2_{33}
$n = 1$	0.0442	0.99906	0.0856	0.99410
$n = 2$	1.0632	0.999983	1.0890	0.999317
$n = 3$	2.0729	0.999986	2.1455	0.999504

(111)

The corresponding innermost-pair exponents for $n = 1, 2, 3$ are $(0.0420, 1.0539, 2.0561)$ for $(2,2)$ and $(0.0745, 1.0278, 2.0427)$ for $(3,3)$. The cubic rescaled

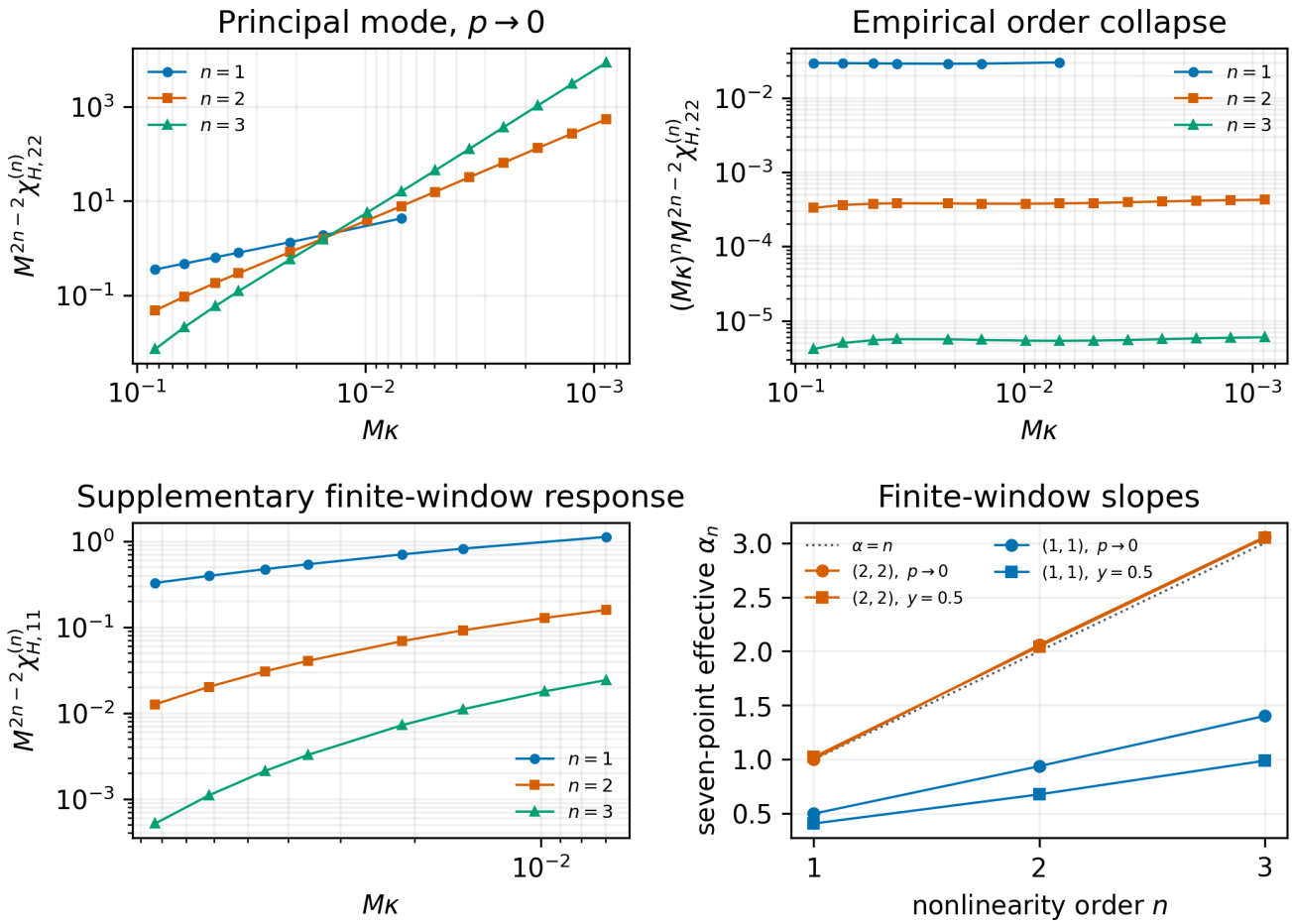


FIG. 7. Operator-order dependence of the near-extremal response. (a) The $(2,2)$ threshold response for $n = 1, 2, 3$. (b) Multiplication by $(M\kappa)^n$ produces a plateau for the principal mode. (c) The supplementary $(1,1)$ response grows with n but does not obey the principal collapse. (d) Seven-point effective slopes at the threshold and at fixed $y = 0.5$; the dotted line is $\alpha_n = n$. The supplementary slopes are finite-window diagnostics. Only paired-resolution accepted points enter the curves and fits.

weights vary by only 2.7% and 5.2% across their three-spin intervals, and by 1.3% and 2.3% across the innermost pairs. For $n = 2, 3$, the corresponding full-interval variations are 20.2%, 27.0%, 23.1%, and 42.9%, falling to 7.4%, 3.8%, 7.7%, and 5.9% on the innermost pairs. The inward movement toward $n - 1$ is more informative than the three-point R^2: only one residual degree of freedom is available, and the remaining offsets are consistent with the finite-window modulation already seen in the pointwise response. The experiment therefore supplies a direct cubic control and supports the predicted envelope without claiming a pure asymptotic power.

The scaling function also changes shape with operator order. At $a/M = 0.9999$ we evaluate $0.75 \le y \le 1.20$ in steps of 0.05 at $N = q = 288, 320$. A five-point local fit

$$\ln \mathcal{F}_{n,\ell}(y) = \ln \mathcal{F}_{n,\ell}(y_{\rm pk}) - \frac{(y - y_{\rm pk})^2}{2\sigma_y^2} + \cdots \qquad (112)$$

gives

	$(2,2)$			$(3,3)$		
n	1	2	3	1	2	3
$y_{\rm pk}$	0.887	0.980	1.027	0.891	0.976	1.015
σ_y	0.307	0.236	0.205	0.319	0.246	0.231

(113)

All 60 dense-scan rows pass; the largest paired-response change is 1.50×10^{-4} and the largest flux-relative balance residual is 3.42×10^{-9}. Increasing n therefore shifts the peak toward the absorptive side and narrows it within the common $p = O(\kappa)$ frequency scale. The factorization $\widehat{\chi}^{(n)}_H = \mathcal{I}^n_H \mathcal{K}_n$ suggests the local peak condition

$$n\,\partial_y \ln \mathcal{I}_H + \partial_y \ln \mathcal{K}_n = 0. \qquad (114)$$

If the kernel varies more slowly than the repeated intensity factor, increasing n moves the response toward the maximum of $\mathcal{I}_H$ and sharpens it. The measured migration is consistent with this mechanism, but the present

data do not establish a large-n law or an n-independent kernel.

Equation (107) also changes the perturbative control parameter. Defining the dimensionless coupling $\widehat{\epsilon}_n = \epsilon_n/M^{2n-2}$, a principal mode requires

$$g_{\mathrm{eff},n}(y) \equiv |\widehat{\epsilon}_n||\mathcal{A}_{\mathrm{in}}|^{2n}\widehat{\chi}_H^{(n)} \sim \frac{|\widehat{\epsilon}_n||\mathcal{A}_{\mathrm{in}}|^{2n}\mathcal{F}^{\mathrm{num}}_{n,\ell}(y,\ln M\kappa)}{(M\kappa)^n} \ll 1. \tag{115}$$

Consequently the breakdown scale is parametrically $M\kappa \sim [|\widehat{\epsilon}_n||\mathcal{A}_{\mathrm{in}}|^{2n}|\mathcal{F}^{\mathrm{num}}_{n,\ell}|]^{1/n}$ over the resolved principal envelope. If a general $U(1)$-invariant potential contains adjacent nonzero couplings, their principal-mode control parameters obey

$$\frac{g_{\mathrm{eff},n+1}}{g_{\mathrm{eff},n}} \sim \frac{|\widehat{\epsilon}_{n+1}|}{|\widehat{\epsilon}_n|}\frac{|\mathcal{A}_{\mathrm{in}}|^2}{M\kappa}\left|\frac{\mathcal{F}^{\mathrm{num}}_{n+1,\ell}}{\mathcal{F}^{\mathrm{num}}_{n,\ell}}\right|. \tag{116}$$

Thus extremality can reverse the coefficient-defined ordering of adjacent local operators. This does not justify extrapolating to arbitrarily large n: it says instead that no fixed finite truncation is uniform once the right-hand side of Eq. (116) approaches unity. This is the opposite of the monopole result: the same $n > 1$ operator is irrelevant as $\omega \to 0$ in the axisymmetric far zone, but increasingly relevant in a principal near-superradiant throat mode. It is a statement about first-order susceptibility, not a computation of the nonlinear state beyond the breakdown scale.

The $\ell = m = 0$ mode is a stronger control because it also tests the lowest scalar multipole. Here $p = \omega$ and the natural scan coordinate is $y = \omega/\kappa$. We computed $0.0125 \le y \le 8$ at $a/M = 0, 0.5, 0.9, 0.98, 0.99, 0.995, 0.997, 0.999$ and performed quadratic $y \to 0$ extrapolations over the stable low-frequency windows. Table XIX compares the extrapolations with Eq. (64). The largest relative discrepancy is 6.6×10^{-5} and the largest primary two-resolution change is 2.0×10^{-4}; the largest linear-balance and Wronskian errors in the fit samples are 1.7×10^{-12} and 4.3×10^{-15}. The response remains bounded while $M\kappa\chi_{H,00}$ decreases as $a/M \to 1$. Thus the inverse-surface-gravity growth is neither a generic property of $s = 0$ nor an artifact of selecting low ℓ.

At the smallest frequencies the direct reflected-channel interference involves subtraction of nearly equal terms and its separate first-order balance can reach 1.2×10^{-4}. Those rows are not used for $R^{(1)}$ or in the flux spectra. The monopole extrapolation instead uses the independently converged horizon ratio $\chi_H = 2\,\mathrm{Re}\,A_H^{(1)}/|B^{\mathrm{inc}}|^2$, together with the linear-balance, Wronskian, and two-resolution checks quoted above.

D. Kerr-to-Schwarzschild continuity

As a limiting-case check independent of external reference calculations, we hold $(\ell, m, M\omega) = (2, 0, 0.52)$ fixed and sweep the dimensionless spin from $a/M = 0$ to 0.5. The scalar spheroidal eigenvalue, the covariant source projection, the scattering amplitudes, and both flux coefficients are recomputed at every spin. At $a = 0$ the angular basis becomes spherical and the endpoint formulation reduces to the Schwarzschild spectral route; no radial endpoint is removed or replaced by a finite horizon cutoff. Figure 10 shows that $T^{(1)}$ and $-R^{(1)}$ vary smoothly from the $a = 0$ value, while Table XX records the balance and Wronskian diagnostics for the same points. The largest absolute first-order balance residual in this scan is 1.4×10^{-12} and the largest linear balance residual is below 1.2×10^{-11}. This experiment is a continuity and implementation check; it is not used as an external reference result or as a claim that the Kerr and Schwarzschild coefficient conventions are identical without the conversion in Appendix A.

V. DISCUSSION AND OUTLOOK

We have formulated the weakly nonlinear scattering problem for a scalar field on Kerr while retaining the usual incident, reflected, and horizon-flux observables. The linear solution is normalized at the horizon, the incident normalization is restored through B^{inc}, and the nonlinear corrections are read from the interference terms in the physical energy fluxes. In this form the two useful balance laws, $T^{(0)} + R^{(0)} = 1$ and $T^{(1)} + R^{(1)} = 0$, remain the main global checks of the calculation. The extension is not a formal replacement of the Schwarzschild radial variable by a Kerr one: the horizon frequency $p = \omega - m\Omega_H$, spheroidal angular basis, and Σ-weighted nonlinear source all enter the normalization of the observable coefficients.

For the axisymmetric Kerr sample $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$, the present numerical diagnostics support the following qualitative picture. The nonlinear transmission coefficient develops a single dominant Breit–Wigner-type peak for each ℓ, with the peak position lying on the frequency scale of the corresponding scalar Kerr quasinormal mode. The high-frequency tail is well fit by an exponential form in the computed window; AICc and leave-one-out tests prefer it to a power law for all three multipoles. The fitted temperature is of the same order as the Kerr Hawking temperature, although fixing the slope exactly to $-1/T_H$ worsens the residual. On the accepted low-frequency side, the globally balanced plotting windows alone do not justify asymptotic fits. Matched static/far-zone analysis nevertheless fixes the true cubic limit: for every axisymmetric self channel, $T^{(0)}$ and $T^{(1)}$ share the power $(M\omega)^{2\ell+2}$ and their ratio approaches the spin-independent Bessel integral in Eq. (72). Dedicated horizon-side solves verify this statement for $\ell = 0, 1, 2$ below the plotting windows. The monopole integral can be evaluated exactly and gives $\rho_{H,00}(0) = \ln 2/(2\pi) + \mathrm{i}/4$ and $\chi_{H,00}(0) = \ln 2/\pi$; the eight-spin experiment independently verifies its spin independence. The general

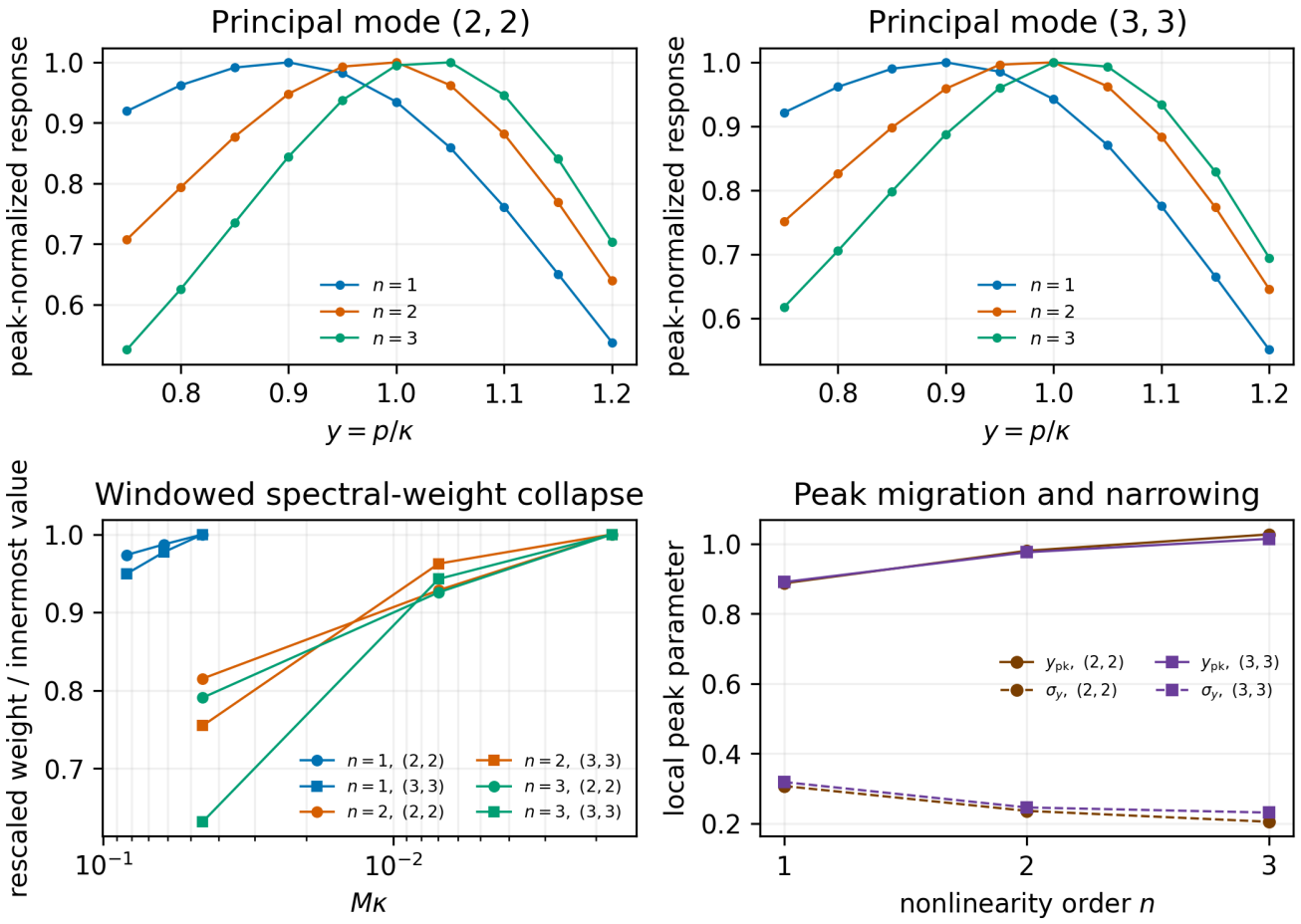


FIG. 8. Finite-bandwidth and line-shape audit for the principal modes. (a),(b) Peak-normalized dense profiles at $a/M = 0.9999$; increasing the local power moves the maximum toward the absorptive side and narrows it while retaining the throat scale $p = O(\kappa)$. (c) The ten-point cubic, quintic, and septic windowed weights, after multiplication by $(M\kappa)^{n-1}$ and normalization to their innermost-spin values. (d) Five-point log-quadratic peak locations and local widths. Every curve uses the higher member of a paired spectral-resolution calculation, and all displayed rows pass the flux and Wronskian gates.

$|\Phi|^{2n}\Phi$ monopole calculation turns this control into an infrared classification: $n = 1$ is marginal, while $n > 1$ responses vanish as ω^{2n-2}. The $n = 2, 3$ double-resolution experiments recover both the integer hierarchy and the analytic coefficients, and their fixed ω/κ scans show corresponding κ^2 and κ^4 suppression toward extremality.

The uniform acceptance cut removes cancellation-limited rows from the globally balanced curves and all two-boundary fits. The separate open infrared points are retained only after paired horizon-response and Wronskian tests. At the lowest filled points, the dedicated scan changes either $T^{(1)}$ or $R^{(1)}$ by at most 2.0×10^{-8} and preserves first-order balance below 3.1×10^{-12}. The numerical evidence therefore supports a resolved decrease on the displayed low-frequency side. Its distinct role is to validate both boundary fluxes; the infrared indices are supplied by Eqs. (68) and (72), whose deeper horizon-side checks do not require a cancellation-prone reflected interference term. The high-frequency result has a different limiting error: the compact inner Green integral requires substantially higher quadrature order than the homogeneous spectral solve. The $q = 1792$–2048 audit bounds the representative relative changes in $T^{(1)}$ by 2.5×10^{-4}, while lowering the Chebyshev order at $q = 2048$ changes those values by at most 9.5×10^{-8}. The representative target-channel experiment in Table XIII demonstrates the matrix structure generated by the Kerr source projection and shows that the dominant reflected off-diagonal amplitudes can be resolved at the stated spectral order. The nonaxisymmetric survey additionally shows that $T^{(1)}$ vanishes linearly with p while $\chi_H = T^{(1)}/T^{(0)}$ has a finite positive threshold limit at every fixed subextremal spin. The extended scan reveals that this limit is not uniform as $a/M \to 1$: the principal $\ell = m = 2$–5 responses grow approximately as $1/\kappa$ along the tested paths, the $\ell = m = 2, 3$ curves collapse in the near-horizon variables of Eq. (91), and the multipole hierarchy reverses. The real-weight $(1, 1), (2, 1), (3, 2)$ controls do not share

TABLE XIX. Zero-frequency scalar-monopole response. The second column is a quadratic extrapolation in $y = \omega/\kappa$; the third compares it with $\ln 2/\pi$, and the last gives the largest relative change between the two radial resolutions in the fit window. The fits use $y = (0.0125, 0.025, 0.05)$ except at $a/M = 0.999$, where the stable $y = (0.05, 0.1, 0.2)$ window is used.

a/M	$\chi_{H,00}(0)$	relative analytic error	maximum resolution change
0.000	0.2206211	6.57×10^{-5}	1.46×10^{-8}
0.500	0.2206249	4.85×10^{-5}	1.22×10^{-8}
0.900	0.2206330	1.20×10^{-5}	6.83×10^{-7}
0.980	0.2206335	9.60×10^{-6}	1.84×10^{-7}
0.990	0.2206351	2.18×10^{-6}	6.21×10^{-8}
0.995	0.2206353	1.23×10^{-6}	1.63×10^{-7}
0.997	0.2206354	8.24×10^{-7}	4.10×10^{-7}
0.999	0.2206343	5.72×10^{-6}	1.97×10^{-4}

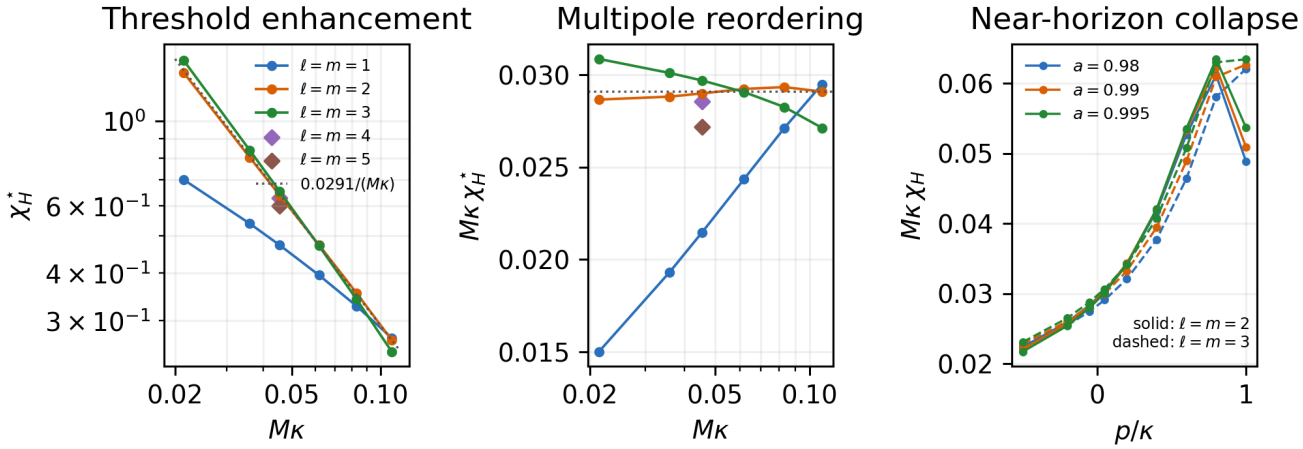


FIG. 9. Near-extremal nonlinear response. (a) The raw threshold response grows as the surface gravity decreases; solid guides show power-law fits. (b) Multiplication by $M\kappa$ exposes the reversal of the low-multipole hierarchy and the approximately constant $\ell = m = 2, 3$ scale. Open points at $a/M = 0.995$ are the $\ell = m = 4, 5$ checks. (c) The full $\ell = m = 2$ (solid) and 3 (dashed) response approximately collapses when plotted as $\Xi = M\kappa\chi_H$ against $y = p/\kappa$. Every displayed point passes the two-resolution response, flux-balance, and Wronskian gates.

this scaling. For the higher $|\Phi|^{2n}\Phi$ interactions, the principal $(2, 2)$ and $(3, 3)$ channels instead display the numerical order-locked envelope $M^{2n-2}\chi_H^{(n)} = (M\kappa)^{-n}\mathcal{F}^{\rm num}_{n,\ell}$ for $n = 2, 3$, both at threshold and at fixed $p/\kappa = 0.5$. Seven-point free fits resolve small residual slopes in $\mathcal{F}^{\rm num}_{n,\ell}$, so this statement is an order hierarchy rather than an exact pure-power identity. The low-real-weight $(1, 1)$ channel acquires weaker order-dependent growth, while the accepted $(2, 1)$ window remains a weak crossover. This separates the operator relevance of the throat from the infrared irrelevance of the same higher powers in the scalar monopole. The finite-bandwidth experiment sharpens that distinction. Integrating over a fixed p/κ interval changes the pointwise κ^{-n} hierarchy to κ^{1-n} rather than removing it. Moreover, the dense profiles show a systematic drift of $y_{\rm pk}$ toward the absorptive side and a decreasing local width as n rises from 1 to 3. The thermal scale is therefore common, but the scaling function is operator dependent: higher local powers amplify and concentrate a different part of the same throat-frequency window. For the signed coupling $\epsilon > 0$ this enhances superradiant amplification; the conventional positive quartic potential, $\epsilon = -\lambda$, suppresses it at first order.

A. Physical content of the Kerr extension

The genuinely Kerr-specific result is not merely a displacement of the axisymmetric line shapes. The explicit factor $p = \omega - m\Omega_H$ forces both the linear and first-order horizon fluxes to vanish at the superradiant threshold, while the independently evaluated horizon amplitude shows that their ratio remains regular at fixed subextremal spin. For every accepted family we find

$$\chi_H^\star > 0, \qquad \operatorname{sgn} T^{(1)} = \operatorname{sgn} p, \qquad R^{(1)} = -T^{(1)}. \quad (117)$$

Thus the threshold zero is a physical flux zero rather than a singularity in p. Equation (117), together with

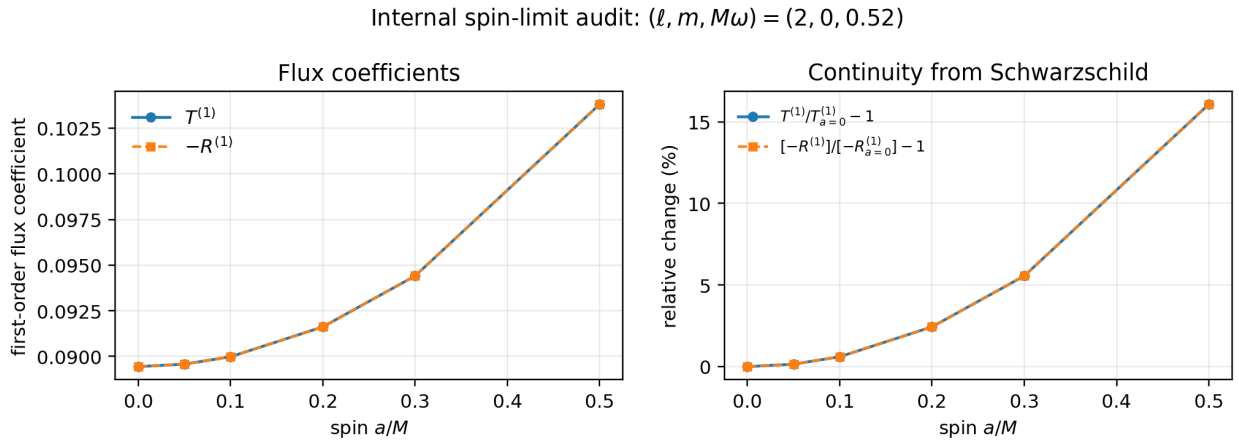


FIG. 10. Internal Kerr-to-Schwarzschild continuity test at fixed $(\ell, m, M\omega) = (2, 0, 0.52)$. The first-order transmission coefficient and $-R^{(1)}$ remain paired by the flux balance while changing smoothly with a/M. The right panel normalizes both coefficients to the $a = 0$ value; all points use the endpoint-inclusive spectral solver and are accepted by the first-order balance check.

TABLE XX. Internal Kerr-to-Schwarzschild spin-limit audit for $(\ell, m, M\omega) = (2, 0, 0.52)$. The entries use the same fixed-incident normalization as the production coefficients. The two balance columns are $T^{(0)} + R^{(0)} - 1$ and $T^{(1)} + R^{(1)}$, respectively.

a/M	$T^{(1)}$	$R^{(1)}$	$\lvert\delta_0\rvert$	$\lvert\delta_1\rvert$	$\lvert\delta_W\rvert$
0.00	8.94395×10^{-2}	-8.94395×10^{-2}	1.1×10^{-12}	3.8×10^{-13}	2.6×10^{-16}
0.05	8.95742×10^{-2}	-8.95742×10^{-2}	9.8×10^{-12}	1.1×10^{-12}	3.3×10^{-16}
0.10	8.99794×10^{-2}	-8.99794×10^{-2}	3.7×10^{-12}	4.9×10^{-13}	0
0.20	9.16178×10^{-2}	-9.16178×10^{-2}	4.3×10^{-13}	7.6×10^{-14}	1.8×10^{-16}
0.30	9.44095×10^{-2}	-9.44095×10^{-2}	1.1×10^{-11}	1.3×10^{-12}	1.8×10^{-16}
0.50	1.03814×10^{-1}	-1.03814×10^{-1}	6.0×10^{-12}	1.2×10^{-12}	0

Eq. (49), gives a direct energy-flux statement: for $p < 0$ the coefficient $T^{(1)}$ is negative and the reflected coefficient is positive. The signed interaction $\epsilon > 0$ therefore increases the superradiant excess $Z = R - 1$, while the standard positive quartic potential, whose action convention gives $\epsilon = -\lambda$, reduces it at first order.

The new information is that regularity in p is not uniform in surface gravity. For $\ell = m = 2$, Eq. (97) changes the fractional correction in Eq. (49) into the effective near-extremal parameter

$$g_{\rm eff} \equiv |\epsilon||\mathcal{A}_{\rm in}|^2\chi_H \sim \frac{C|\epsilon||\mathcal{A}_{\rm in}|^2}{M\kappa}, \qquad C \simeq 0.03. \tag{118}$$

The first-order expansion consequently requires

$$|\epsilon||\mathcal{A}_{\rm in}|^2 \ll \frac{M\kappa}{C} \tag{119}$$

in the enhanced principal sector. Therefore a coupling and incident amplitude that are perturbatively small at moderate spin need not remain so as $a/M \to 1$. This is a physical breakdown criterion for the weakly nonlinear expansion, not a computed saturation mechanism. Determining the subsequent behavior requires a resummation or a fully nonlinear evolution.

For the general interaction, Eq. (115) replaces the cubic criterion by

$$|\widehat{\epsilon}_n||\mathcal{A}_{\rm in}|^{2n} \ll \frac{(M\kappa)^n}{|\mathcal{F}^{\rm num}_{n,\ell}(y, \ln M\kappa)|} \tag{120}$$

in a principal mode over the resolved envelope. The numerical order locking therefore describes a hierarchy of increasingly singular finite-window susceptibilities, even though the fitted plateau values decrease with n. Equation (116) makes the implication for a multi-operator potential explicit: at fixed incident amplitude, each successive order receives one additional throat factor $1/(M\kappa)$, so the moderate-spin operator ordering can be inverted before the first-order series loses control. The crossover depends on the coupling ratios and scaling functions and is not universal. Its exact-extremal continuation remains conditional on the nonlinear kernel. No comparable universal law is claimed for supplementary modes.

The physical picture supported by the factorization is a principal Kerr throat acting as a fixed-flux intensity amplifier. The horizon energy flux remains regular and carries $p \sim \kappa$, while the principal fixed-incident field intensity in Eq. (102) has a κ^{-1} envelope. A local interaction samples n powers of that intensity, but the unit-horizon nonlinear kernel can modify the residual scaling.

Equations (100)– (104) therefore isolate the analytically known linear contribution without pretending to calculate the nonlinear one. The observed κ^{-n} envelope is the numerical statement that their product has no additional order-sized power on the principal paths; the small residual slopes in Table XVIII remain resolved.

The collapse in Eq. (98), the conformal-class split in Table XV, and the multipole reordering all point to a throat-controlled response. In particular, the exceptional $(1,1)$ result shows that corotation alone is insufficient: the robust inverse-κ numerical envelope appears only after the conformal weight becomes complex. This interpretation is consistent with the linear Green-function branch point at the extremal superradiant bound and with the self-similar near-horizon sector [47–50], and the fixed-y variables agree with the thermal Kerr/CFT description of near-superradiant scattering [42, 43]. None of those linear results derives the exponent or coefficient of χ_H without a separate statement about $\mathcal{K}_{n,h}$. Moreover, the complex principal weight allows bounded log-periodic phases in $\ln\kappa$; the present effective-slope fits establish the leading envelope over the resolved spins, not the absence of such phases. The windowed audit in Eq. (109) verifies the corresponding finite-bandwidth hierarchy directly: a thermal-width integration removes one inverse power, leaving the cubic spectral weight finite at leading power but the $n = 2, 3$ weights enhanced as κ^{-1} and κ^{-2}. The dense scan further shows that higher local powers reshape the throat line profile, moving its maximum toward the absorptive side and narrowing it. These are properties of integrated monochromatic susceptibilities, not a substitute for the frequency mixing in a nonlinear wave packet. The axisymmetric infrared result sharpens this conclusion: every cubic $(\ell, 0)$ self-channel ratio remains finite and spin independent as $\omega \to 0$, with the first three constants listed in Table I. Hence $M\kappa\chi_{H,\ell 0} \to 0$ rather than approaching the nonzero scaling function seen for the principal $\ell = m \geq 2$ modes. The near-extremal enhancement is therefore tied to the principal corotating throat sector, not to a universal infrared growth of the nonlinear scalar response. Equation (85) strengthens this statement beyond the quartic potential within the monopole: at fixed ω/κ, every higher $U(1)$-invariant local operator tested there is suppressed by an additional even power of surface gravity. The scalar monopole therefore defines an infrared universality hierarchy in which the cubic equation is the unique marginal member of the family, rather than the first example of a generic near-extremal divergence. In contrast, Eq. (107) shows that the principal nonaxisymmetric throat sector assigns the same operators the growing hierarchy κ^{-n}. The far-zone and throat limits thus give opposite relevance classifications for one and the same local interaction. At every fixed subextremal spin $\chi_H^\star$ is finite, whereas the sequence of threshold limits grows without a resolved plateau for $\ell = m = 2, 3$. The calculation therefore establishes a nonuniform approach to extremality and strongly motivates a separate analysis of the order of the $p \to 0$ and $a/M \to 1$ limits; it does not numerically impose $a = M$.

These statements use only the continuous frequency intervals satisfying the two-resolution response, linear-flux, and Wronskian gates. Rejected low-frequency islands and cancellation-limited high-frequency tails are not used. Where the direct reflected interference term fails its separate balance gate, only the independently converged horizon-side quantities $T^{(1)}$ and χ_H enter the physical conclusions.

The spin-limit audit supplies an additional implementation-level check. We fix $(\ell, m) = (2, 0)$ and $M\omega = 0.52$; the first-order coefficients change smoothly from the $a = 0$ endpoint to the production value at $a = 0.5$, while the two flux-balance laws remain satisfied. This continuity test does not replace a convention-matched Schwarzschild comparison, but it rules out a discontinuity introduced solely by the Kerr tortoise coordinate, spheroidal projection, or endpoint matching.

B. Relation to previous nonlinear calculations

The boundary-value problem is the decisive distinction from previous self-interacting Kerr scalar calculations. The axion and boson-cloud studies of Refs. [20, 22–25] evolve massive quasi-bound levels whose amplitudes grow through a complex-frequency superradiant instability. Their nonlinear observables include occupation-number transfer, emitted scalar or gravitational radiation, shifts of the bound-state frequency, and the onset of saturation or collapse. Here ω is real, the field is massless, and an incident beam continuously supplies flux from infinity. Consequently $T^{(1)}$ and $R^{(1)}$ are derivatives of open-channel energy fluxes at fixed incident normalization. They cannot be inserted as growth-rate corrections in a cloud evolution equation without an additional projection onto quasi-bound states.

The closest stationary scalar comparison is the nonlinear Q-cloud of Ref. [21]. Both calculations use a complex field with a $U(1)$-invariant local potential and both single out the synchronization condition $p = \omega - m\Omega_H = 0$. The Q-cloud is nevertheless a massive, spatially bound, finite-amplitude solution with zero horizon flux; the present solution is a perturbative scattering state with oscillatory data at infinity. Our threshold result is therefore not a small-amplitude Q-cloud branch. Rather, it states that the two fluxes vanish linearly in p while their first-order ratio has a finite limit at fixed subextremal spin, followed by a nonuniform $1/\kappa$ enhancement along the near-extremal sequence.

The closest analytic antecedent for the throat scaling is linear Kerr/CFT. Near-superradiant scalar graybody factors and their charged and spinning generalizations are reproduced by thermal two-point functions [42–44], and source-driven radiation from the throat can likewise be related to a boundary transition rate [46]. The present observable goes one step beyond this linear map. The cu-

bic equation comes from a quartic bulk vertex and hence probes a four-point contact response; it is not the same quantity as the bulk cubic vertex and three-point correlator studied in Ref. [45]. Our result supplies the asymptotically flat matching, fixed-incident normalization, and signed flux derivative that those correlators do not determine. Kerr/CFT and matched linear theory therefore fix the intensity factor and organize the conformal-sector split, but do not by themselves determine the exponent, coefficient, or sign of the nonlinear susceptibility. Those are new Green-function data in the present calculation.

The nonlinear gravitational-scattering calculations of Refs. [17–19, 37] are closer in their incoming-versus-outgoing interpretation, but include the metric dynamics and a finite-bandwidth pulse. Their harmonic generation and spectral broadening also explain why the present $U(1)$-invariant, single-frequency response is only one component of a general nonlinear scattering map. Likewise, the Kerr QNM projection and near-extremal amplitude system of Refs. [38, 39] concern complex-frequency gravitational modes and time-dependent nonlinear transfer. The observed $1/\kappa$ scalar susceptibility is consistent with a throat-controlled nonuniform limit, but neither the inverse-cascade solution nor QNM orthogonality yields its coefficient C or its flux sign.

This comparison isolates the new content of the present calculation: a flux-normalized, real-frequency Green-function susceptibility for massless self-interacting scalar scattering on fixed Kerr; its signed continuation through the superradiant threshold; the resolved near-extremal scaling and conformal-class split of the nonaxisymmetric sector; the analytic monopole hierarchy for isolated $|\Phi|^{2n}\Phi$ operators; and the opposite, order-locked relevance hierarchy of those operators in principal throat modes, including its finite-bandwidth continuation and operator-dependent peak migration. It also fixes the interpretation: the calculation is a controlled weak-field response function that can inform, but does not replace, a nonlinear wave-packet evolution or a superradiant-cloud model.

C. Scope and limitations

The conclusions above have a deliberately narrow interpretation. The calculation is a fixed-background scalar response problem with prescribed $U(1)$-invariant local interactions. The $n = 2, 3$ monopole experiments test the analytic far-zone hierarchy, while the four-mode near-extremal experiment tests the distinct throat hierarchy. It is not a second-order metric perturbation, a gravitational self-force calculation, or an EMRI waveform model. The main frequency sweep is axisymmetric, while the nonaxisymmetric extension uses a seven-mode common grid through $a/M = 0.999$, selected cubic channels through $a/M = 0.9999$, and the higher-power principal audit through $a/M = 0.9999984375$, together with a ten-point bandwidth audit through $a/M = 0.99999375$, a dense peak scan at $a/M = 0.9999$, and the $\ell = m = 0$ control. It does not determine the exact extremal coefficient, nonlinear saturation, an analytic conformal-weight dependence, a real-time Kerr/CFT multipoint correlator, or generic mode-coupled superradiance. In particular, a finite limit of the scaled unit-horizon kernel in Eq. (103) is not proved. The principal-sector response collapse shows no additional order-sized power for $n = 1, 2, 3$, while the seven-point fits and separate factorization audit resolve modest finite-window two-branch and kernel drift. No asymptotic supplementary-sector critical exponent has been established. The target-channel experiment establishes the spheroidal coupling structure and the dominant reflected amplitudes, but the $\ell' = 6$ horizon coefficient is conditioning limited in double precision. The QNM comparison uses real-frequency peak locations only, and the finite-window exponential fits should not be interpreted as QNM residues or as a proof of Hawking thermality. Extending the result to generic Kerr orbits or to gravitational perturbations requires, respectively, source-harmonic and orbit-evolution modules or metric reconstruction, gauge completion, and regularization; none of these are hidden in the present scalar result. A real scalar with a Φ^3 source would additionally generate harmonics such as 3ω; the single-frequency closure used here relies on the complex field and its $U(1)$-invariant interaction.

DATA AND CODE AVAILABILITY

The numerical data used in Figs. 2–10, including Figs. 1, 7, and 8, and Tables I, II, V, VI, III, VII, VIII, IX, X, XI, XII, XIII, XIV, XV, XVI, XIX, IV, XVII, XVIII, and XX, together with the source code and scripts required to reproduce them, are contained in a frozen review package. The package is not yet publicly archived because its versioned release is being prepared; during peer review it is available to editors and referees from the authors upon reasonable request. A public archival release with a persistent identifier will be deposited before publication, and the data citation in this statement will be updated to identify that release. The numerical checker compares the rounded entries in Tables V–XIII, the superradiant and near-extremal threshold tables, the scalar-monopole limits, the spin-limit audit, the higher-power modal slopes, bandwidth and peak-migration audits, factorization audit, production-resolution audit, and the low-frequency acceptance boundaries with the tracked CSV files. The artifact manifest records file sizes and SHA-256 hashes for the figure and CSV inputs, including the target-channel values in Table XIII; a human-readable snapshot is included in the review package. No external experimental data are used. In the CSV files, "status=ok" records a completed solver call; the manuscript acceptance decision is a separate, path-dependent postprocessing predicate combining paired-resolution response convergence, linear balance, Wronskian constancy, and, where it is numerically informative,

absolute or flux-relative first-order balance.

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Appendix A: Flux normalizations

1. Projected Kerr source

The scalar wave equation separates only after multiplying by $\Sigma = r^2 + a^2\cos^2\theta$. At first order this gives

$$\Sigma\Box\Phi^{(1)} = -\Sigma|\Phi^{(0)}|^2\Phi^{(0)}. \tag{A1}$$

Using

$$\begin{aligned}\Phi^{(0)} &= e^{-\mathrm{i}\omega t+\mathrm{i}m\phi}S_{\ell m}R_{\ell m\omega},\\ \Phi^{(1)} &= e^{-\mathrm{i}\omega t+\mathrm{i}m\phi}\sum_{\ell'}S_{\ell' m}R^{(1)}_{\ell'|\ell m\omega},\end{aligned} \tag{A2}$$

and projecting with $S^*_{\ell' m}$ gives

$$\mathcal{L}_{\ell' m\omega}R^{(1)}_{\ell'|\ell m\omega} = -\left(r^2C^{(0)}_{\ell'\ell m}+a^2C^{(2)}_{\ell'\ell m}\right)|R^{\mathrm{in}}_{\ell m\omega}|^2R^{\mathrm{in}}_{\ell m\omega}. \tag{A3}$$

Here

$$C^{(j)}_{\ell'\ell m} = \int \mathrm{d}\Omega\,\cos^j\theta\,S^*_{\ell' m}|S_{\ell m}|^2S_{\ell m}, \qquad j=0,2. \tag{A4}$$

This is the point at which Kerr differs from the spherical Schwarzschild reduction: a single incident spheroidal channel sources a matrix of target channels with the same m. Only the target channel $\ell'=\ell$ contributes linearly to the total flux, because only that channel interferes with the zeroth-order outgoing field. Off-diagonal channels contribute to the energy flux first at order ϵ^2.

2. Linear current

The scalar radial current is

$$J[R] = \frac{\Delta}{2\mathrm{i}}\left(R^*\frac{\mathrm{d}R}{\mathrm{d}r}-R\frac{\mathrm{d}R^*}{\mathrm{d}r}\right). \tag{A5}$$

Using the asymptotics in Eq. (22), the far-zone current is

$$J_\infty = -\omega|B^{\mathrm{inc}}|^2+\omega|B^{\mathrm{ref}}|^2. \tag{A6}$$

At the horizon, Eq. (21) gives

$$J_H = -p(r_+^2+a^2). \tag{A7}$$

The equality $J_\infty = J_H$ gives Eq. (25).

3. Fixed-incident nonlinear flux

The Green-function amplitudes $A^{(1)}_{\mathrm{ref}}$ and $A^{(1)}_H$ are computed with the zeroth-order solution normalized to unit horizon amplitude. Scattering coefficients, however, are defined at fixed incident amplitude. Therefore we rescale the zeroth-order solution by

$$\alpha = \frac{1}{B^{\mathrm{inc}}}. \tag{A8}$$

Because the source is cubic, the first-order field scales as $|\alpha|^2\alpha$. The fixed-incident self-channel field has the asymptotic form

$$\begin{aligned}R_{\mathrm{fix}} &\sim \frac{e^{-\mathrm{i}\omega r_*}}{r}\\ &+\left(\frac{B^{\mathrm{ref}}}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_{\mathrm{ref}}}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right)\frac{e^{+\mathrm{i}\omega r_*}}{r}, \qquad r\to\infty,\end{aligned} \tag{A9}$$

and

$$R_{\mathrm{fix}} \sim \left(\frac{1}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_H}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right)e^{-\mathrm{i}pr_*}, \qquad r\to r_+. \tag{A10}$$

The transmitted flux coefficient is therefore

$$\begin{aligned}T &= \frac{p(r_+^2+a^2)}{\omega}\left|\frac{1}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_H}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right|^2+\mathcal{O}(\epsilon^2)\\ &= T^{(0)}+\epsilon\frac{2p(r_+^2+a^2)}{\omega|B^{\mathrm{inc}}|^4}\,\mathrm{Re}\,A^{(1)}_H+\mathcal{O}(\epsilon^2),\end{aligned} \tag{A11}$$

which gives Eq. (45). Similarly,

$$\begin{aligned}R &= \left|\frac{B^{\mathrm{ref}}}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_{\mathrm{ref}}}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right|^2+\mathcal{O}(\epsilon^2)\\ &= R^{(0)}+\epsilon\frac{2\,\mathrm{Re}(\overline{B^{\mathrm{ref}}}\,A^{(1)}_{\mathrm{ref}})}{|B^{\mathrm{inc}}|^4}+\mathcal{O}(\epsilon^2),\end{aligned} \tag{A12}$$

which gives Eq. (46). Conservation of the radial current for the inhomogeneous solution with no incoming first-order radiation then yields the first-order balance law Eq. (52). If the prescribed incident amplitude is $\mathcal{A}_{\mathrm{in}}$ rather than unity, both first-order boundary amplitudes acquire the factor $|\mathcal{A}_{\mathrm{in}}|^2\mathcal{A}_{\mathrm{in}}$ and division by the incident flux leaves the factor $|\mathcal{A}_{\mathrm{in}}|^2$ in Eq. (51).

4. Schwarzschild normalization correspondence

The flux coefficients in this paper use unit incident amplitude. This point matters when comparing them with a reduced Schwarzschild calculation that quotes the first-order Green coefficient with the zeroth-order solution normalized at the horizon. In the nonrotating limit, write the scalar field in the reduced convention

as $\Phi = e^{-\mathrm{i}\omega t} Y_{\ell m} u/r$ and let $\widehat{I}$ and $\widehat{R}$ denote the corresponding incident and reflected amplitudes. The present unit-horizon Teukolsky variable has $R^{\rm in} = r_+ u/r$ at zeroth order, so that

$$B^{\rm inc} = r_+ \widehat{I}, \qquad B^{\rm ref} = r_+ \widehat{R}. \tag{A13}$$

The cubic source is homogeneous of degree three. Consequently the horizon and infinity Green amplitudes obey $A_H^{(1)} = r_+^2 \widehat{A}_H^{(1)}$ and $A_{\rm ref}^{(1)} = r_+^3 \widehat{A}_{\rm ref}^{(1)}$ when the reduced solution is horizon normalized. For $a = 0$ (so $p = \omega$), Eqs. (45) and (46) therefore become

$$T^{(1)}_{\text{unit-inc}} = \frac{2\,\mathrm{Re}\,\widehat{A}_H^{(1)}}{|\widehat{I}|^4}, \tag{A14}$$

$$R^{(1)}_{\text{unit-inc}} = \frac{2\,\mathrm{Re}(\widehat{R}^*\,\widehat{A}_{\rm ref}^{(1)})}{|\widehat{I}|^4}. \tag{A15}$$

If the reduced Schwarzschild coefficients are instead defined with the common unit-horizon convention, $\widehat{T}^{(1)} = 2\,\mathrm{Re}\,\widehat{A}_H^{(1)}/|\widehat{I}|^2$ and analogously for $\widehat{R}^{(1)}$, then

$$T^{(1)}_{\text{unit-inc}} = T^{(0)}\widehat{T}^{(1)}, \qquad R^{(1)}_{\text{unit-inc}} = T^{(0)}\widehat{R}^{(1)}. \tag{A16}$$

Thus a direct comparison of numerical values of $T^{(1)}$ between the two conventions is meaningful only after this normalization conversion. The physical first-order flux correction and the balance law are unchanged.

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